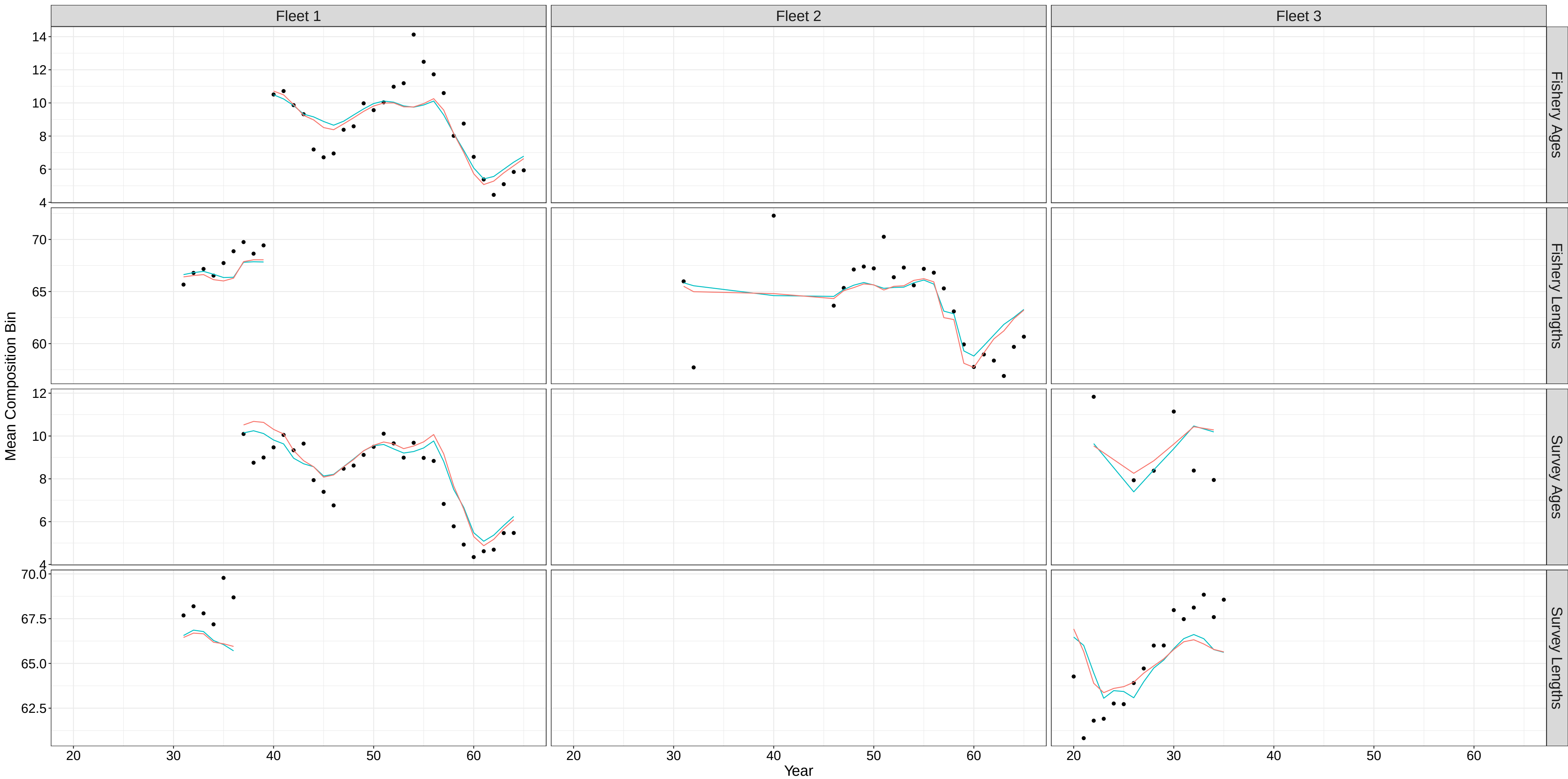


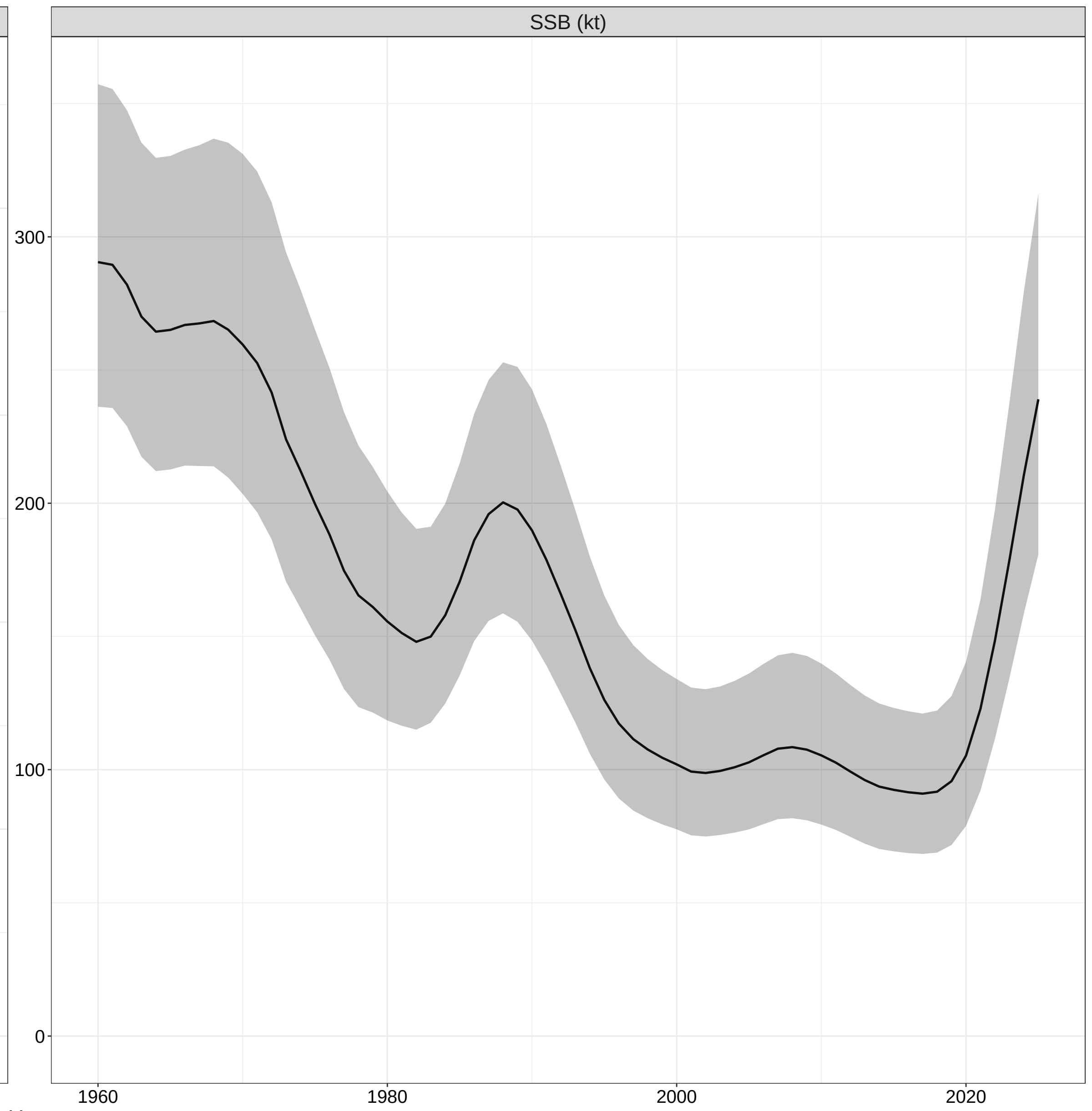
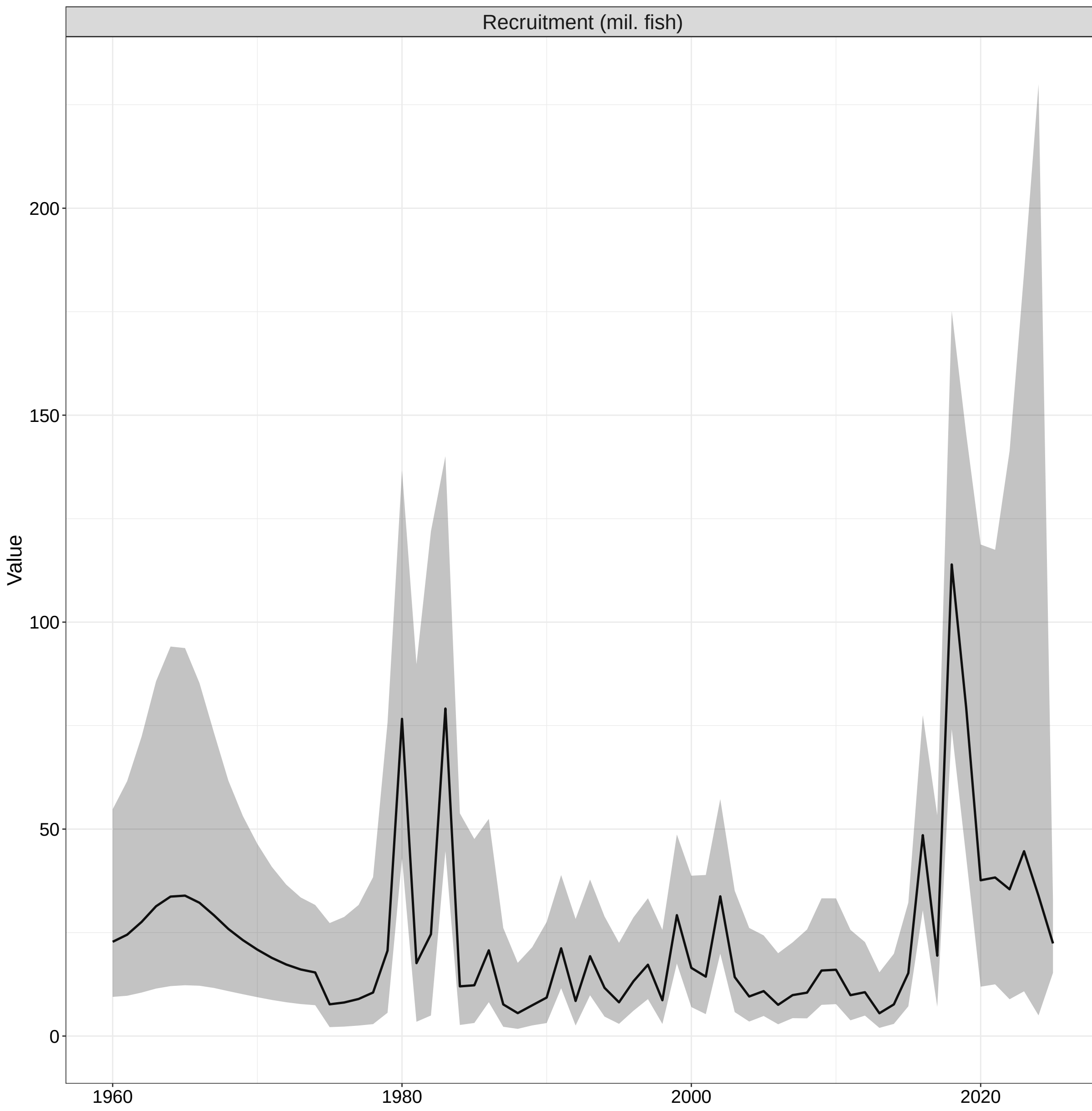
Source	SE	Eff_SE	Data_Wt	ISS	ESS
LL Survey Index	0.07	0.07	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	0.20	0.20	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.12	145.15	16.94
LL Survey Ages	NA	NA	0.38	171.43	65.69
JPN LL Survey Ages	NA	NA	0.14	83.33	11.44
Fixed Gear Fish Lengths	NA	NA	1.00	63.67	63.81
Trawl Fish Lengths	NA	NA	0.23	45.39	10.38
LL Survey Lengths	NA	NA	0.52	133.33	69.27
JPN LL Survey Lengths	NA	NA	0.39	54.62	21.50

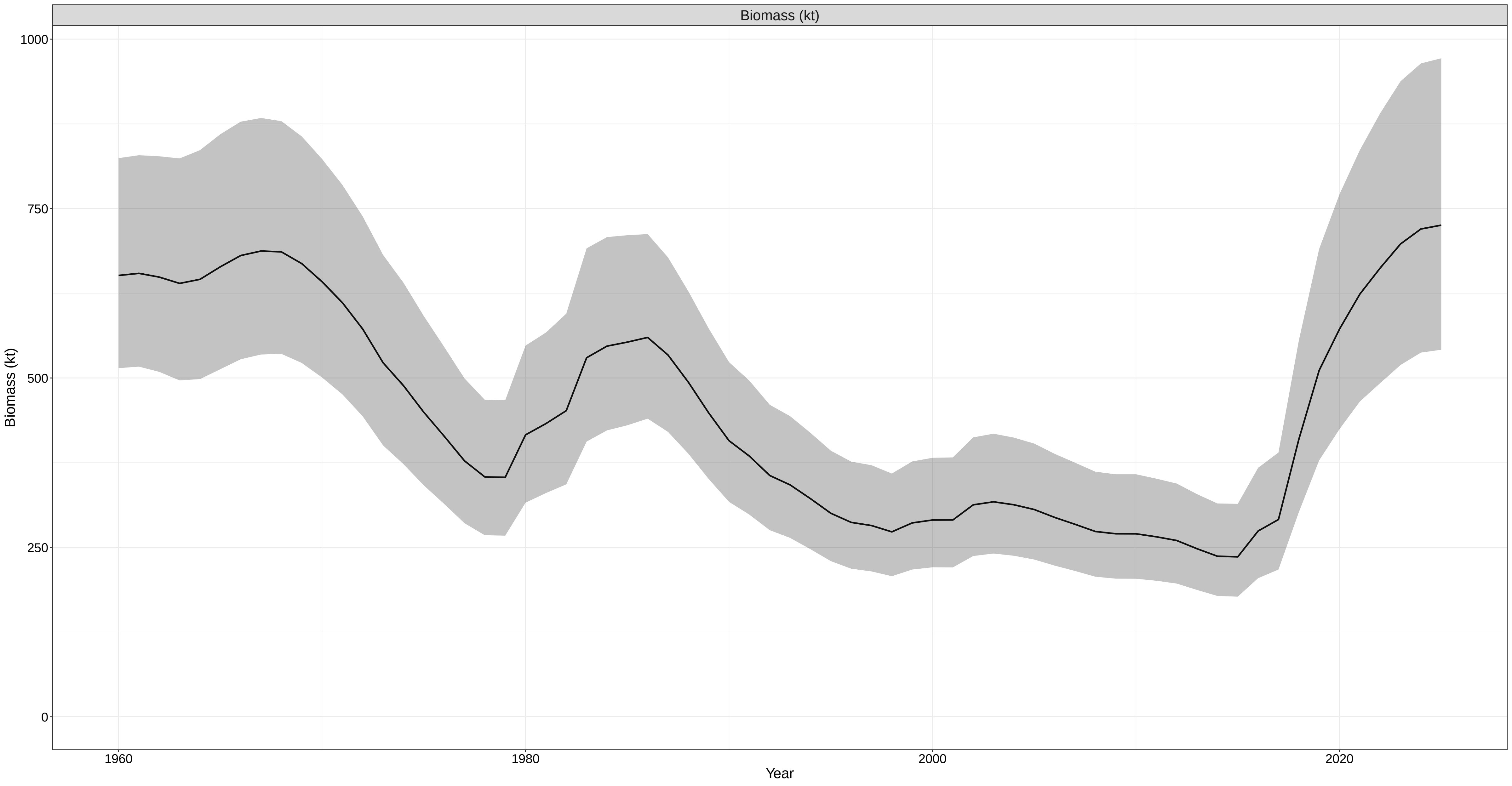
a correlation of 0.9557972540391245. This indicates potential non-convergence according to the tolerance.

colour — First Iter — Last Iter



	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	239.02443	374.65495	0.0405703612077601	49.03881	135.69378	0.0835	1.7615	0.63799	0.48589

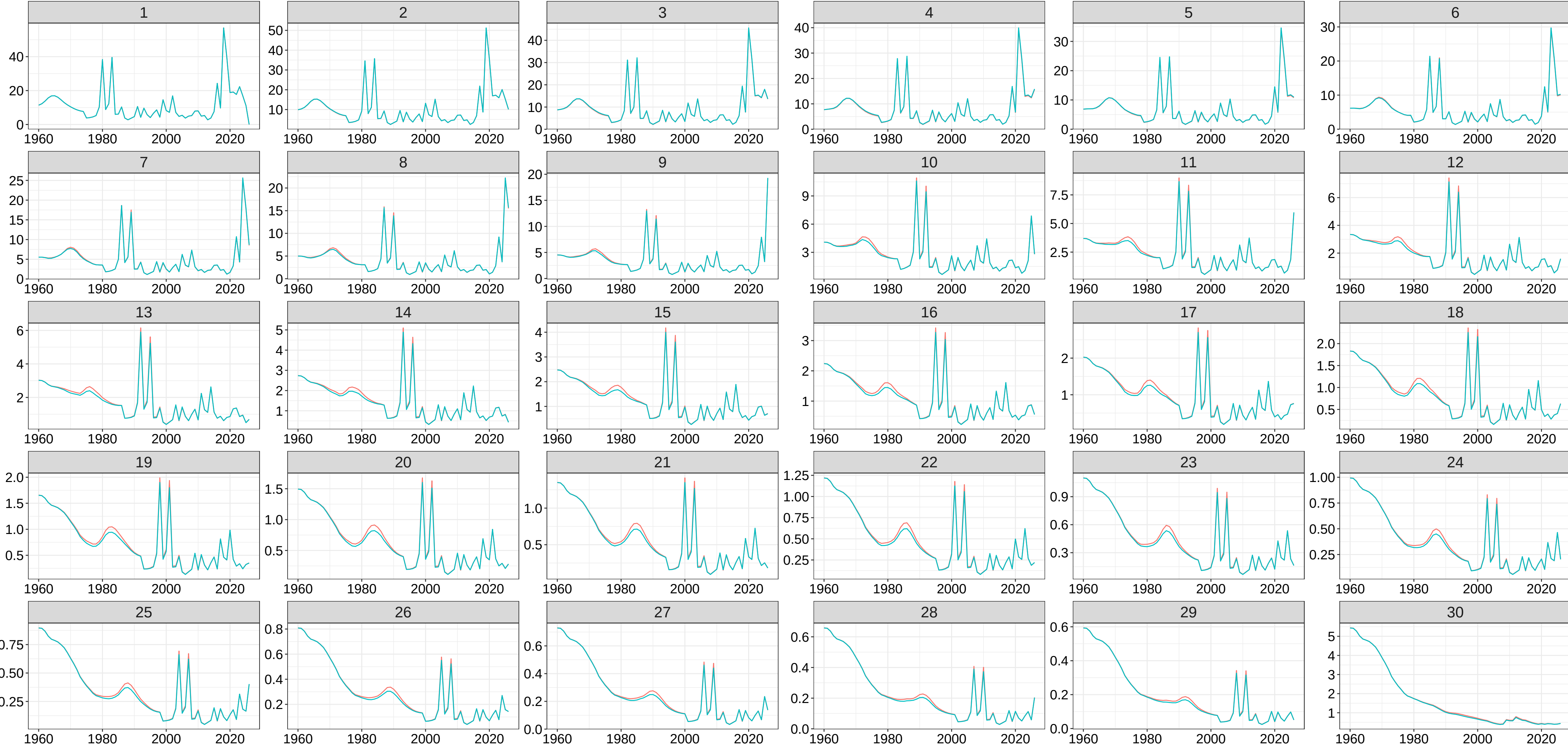




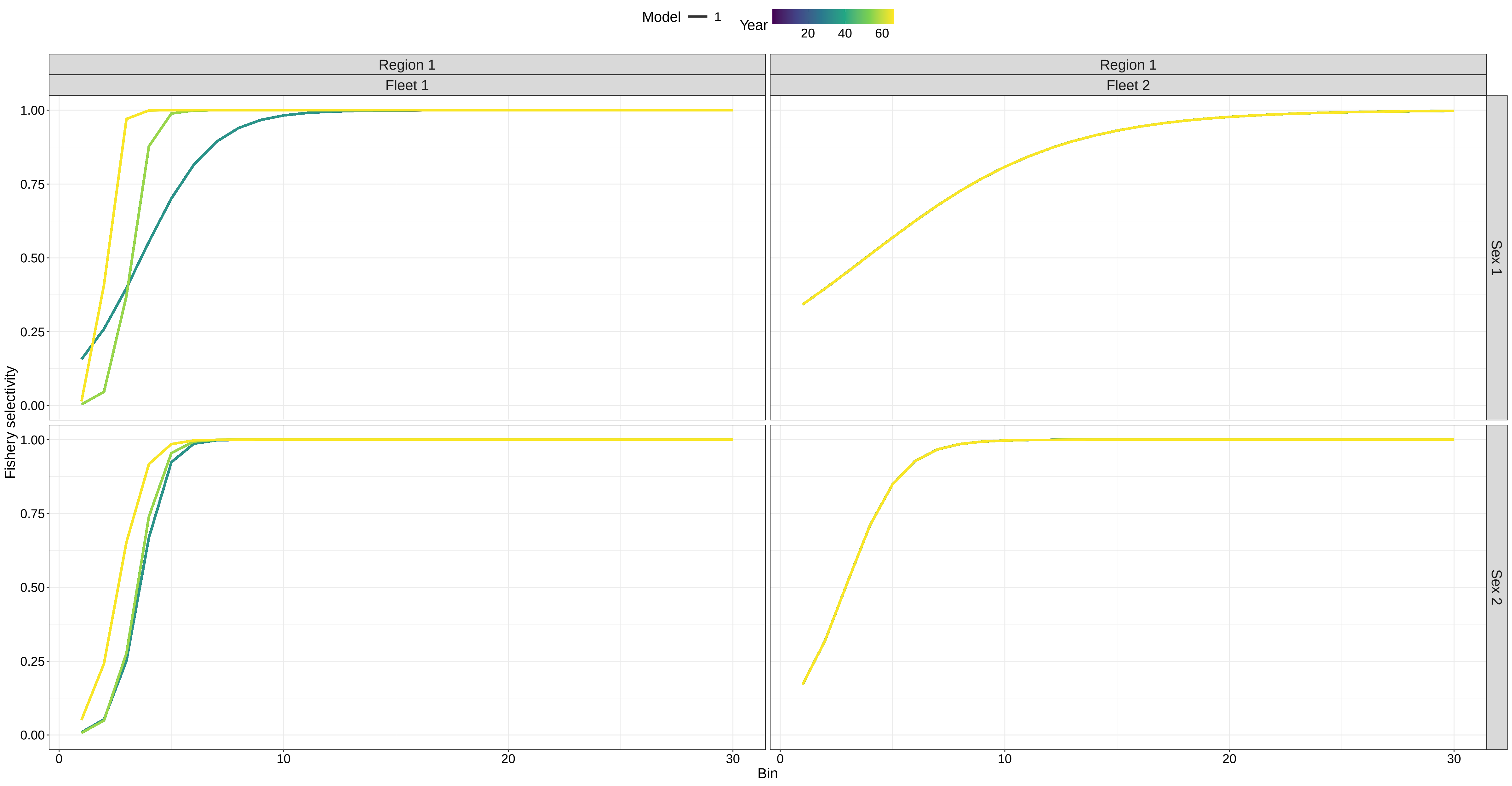
Numbers-at-Age

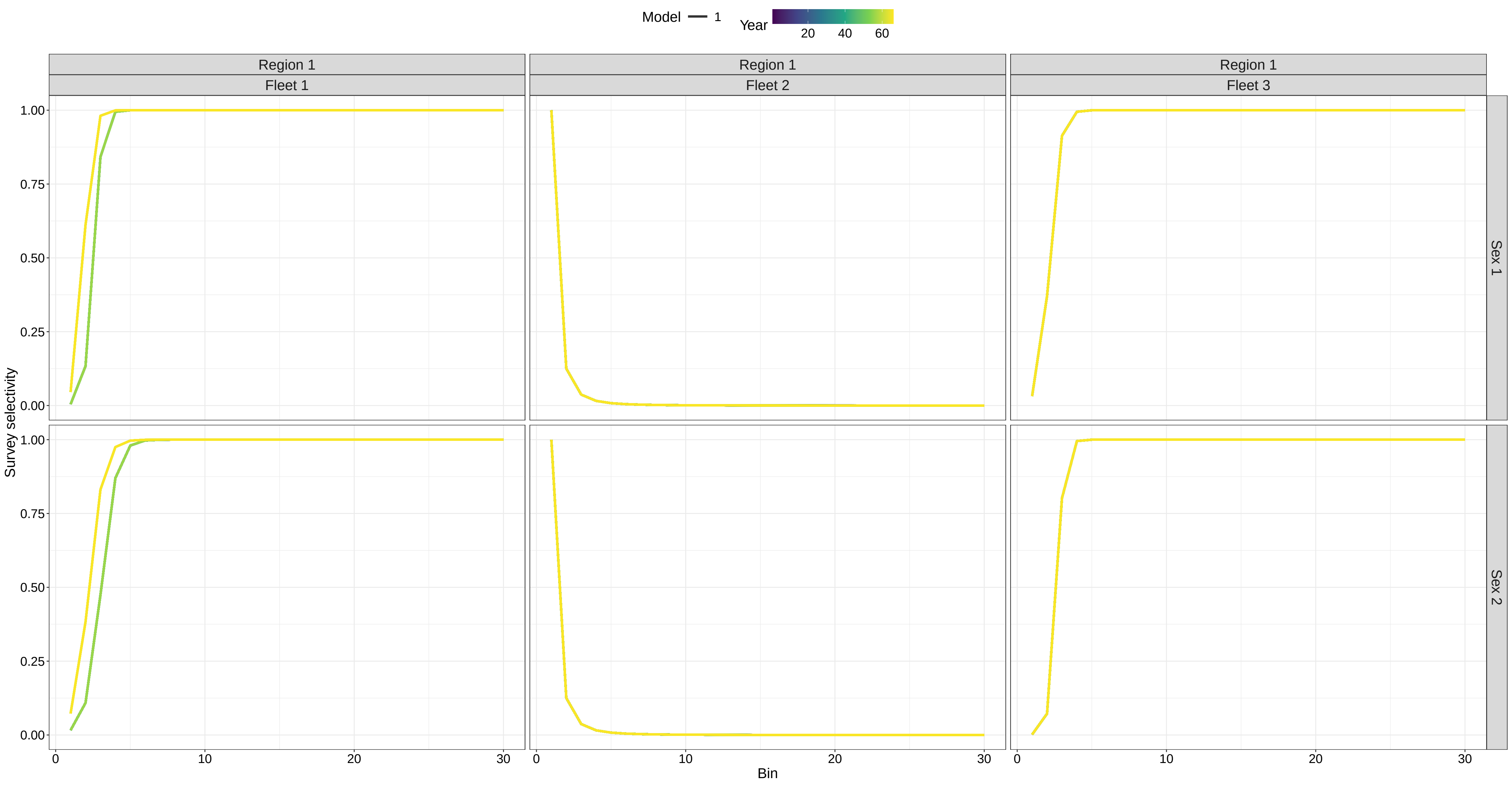
Sex — Female — Male

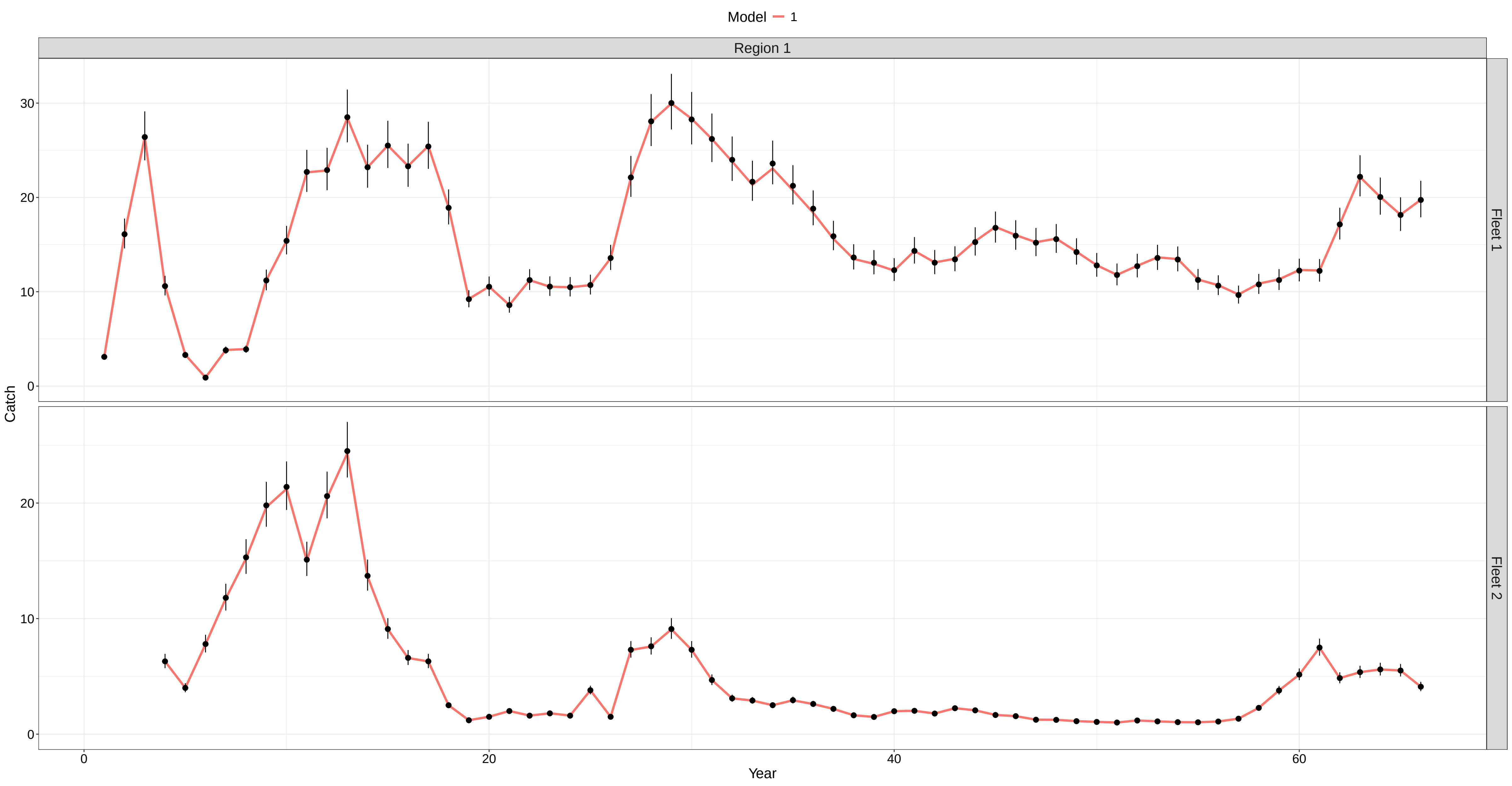
Numbers-at-age (millions)



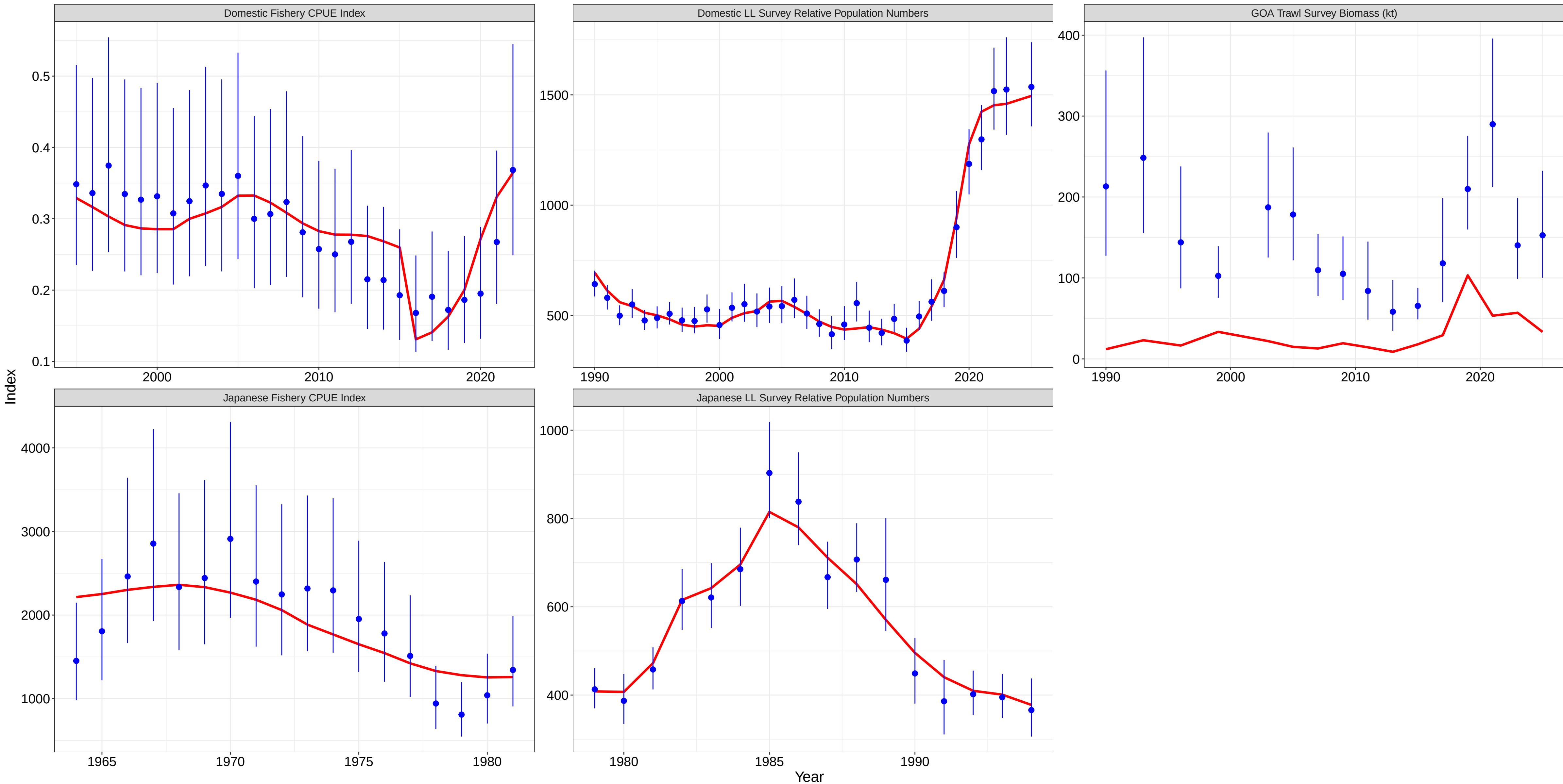
Year



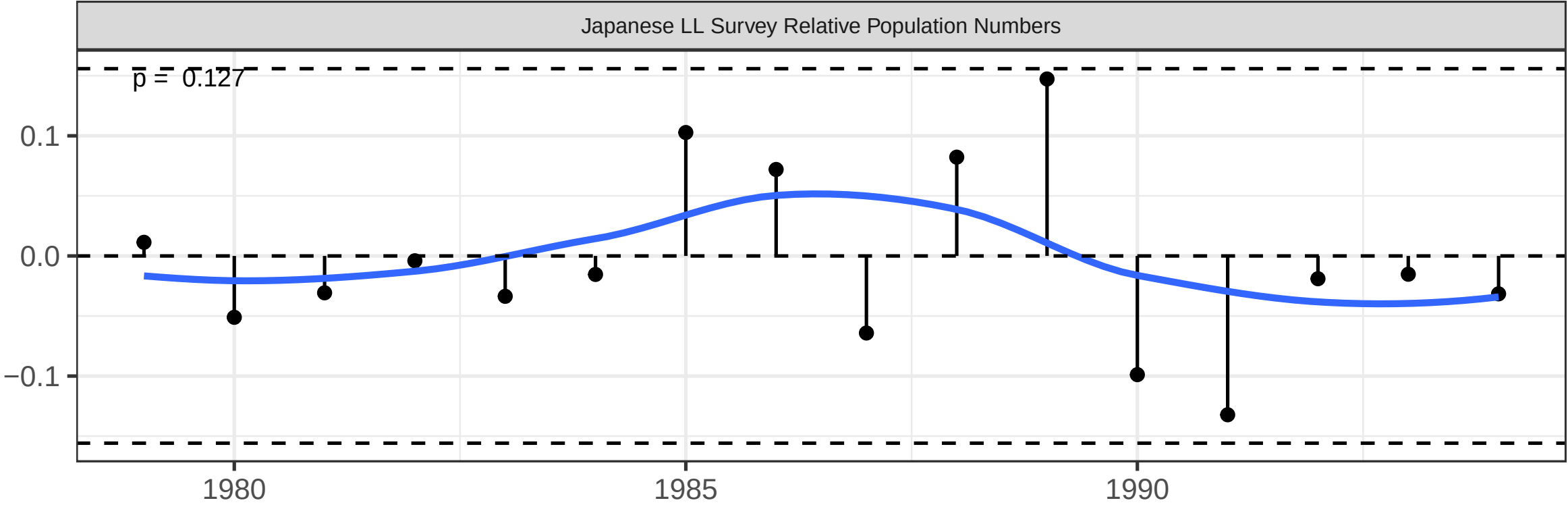
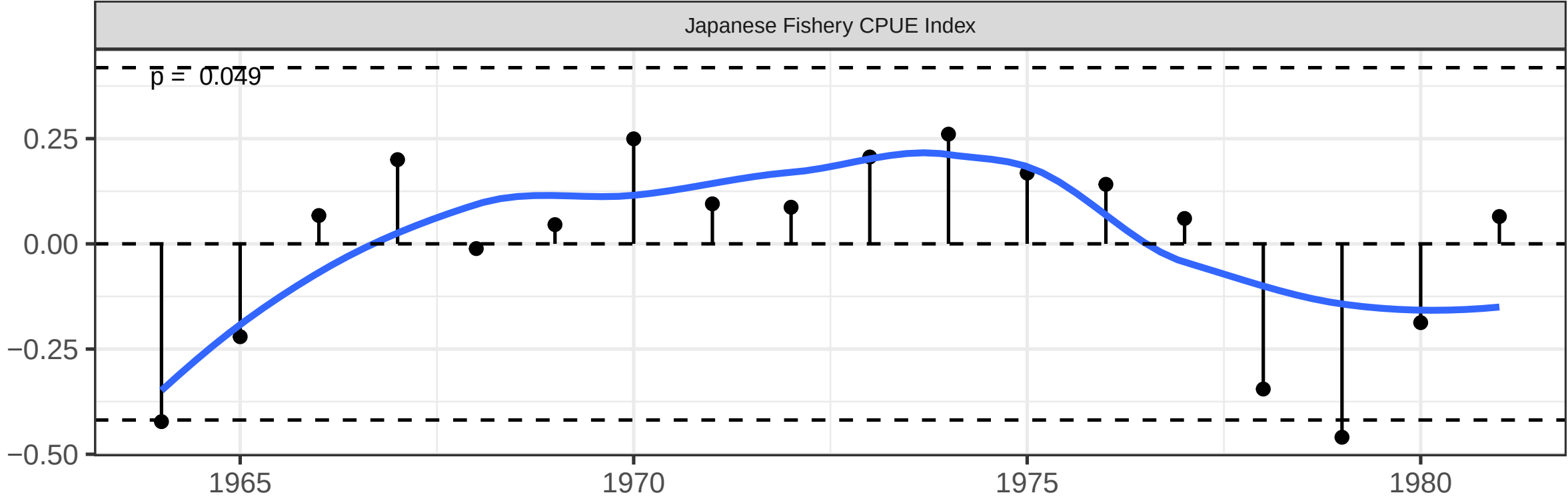
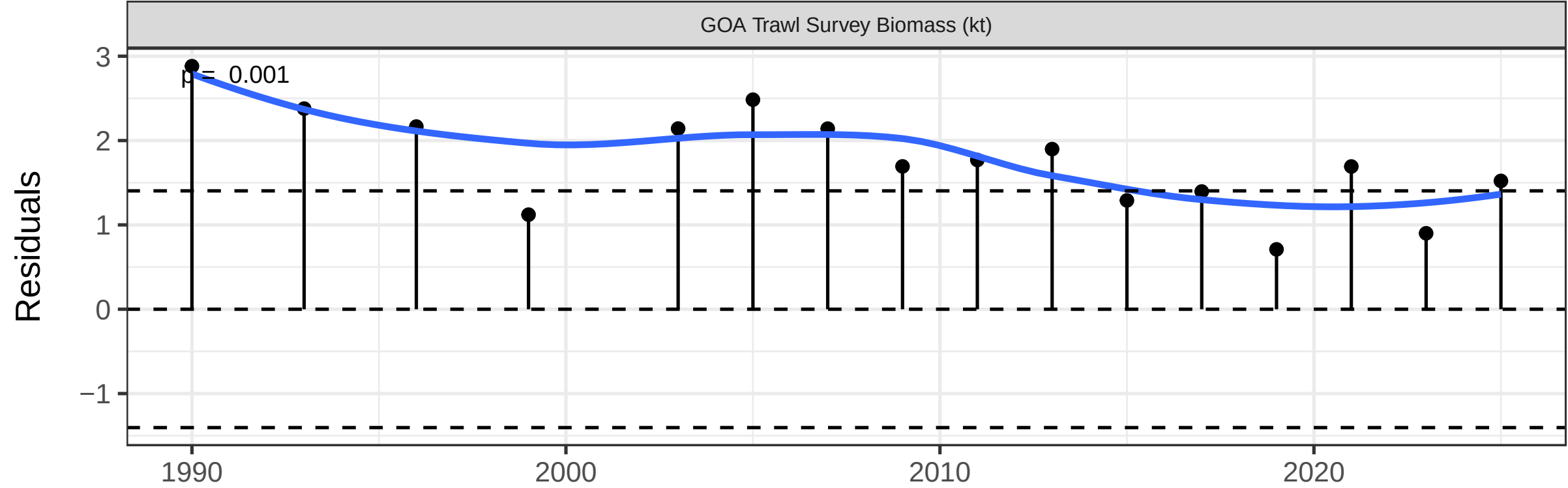
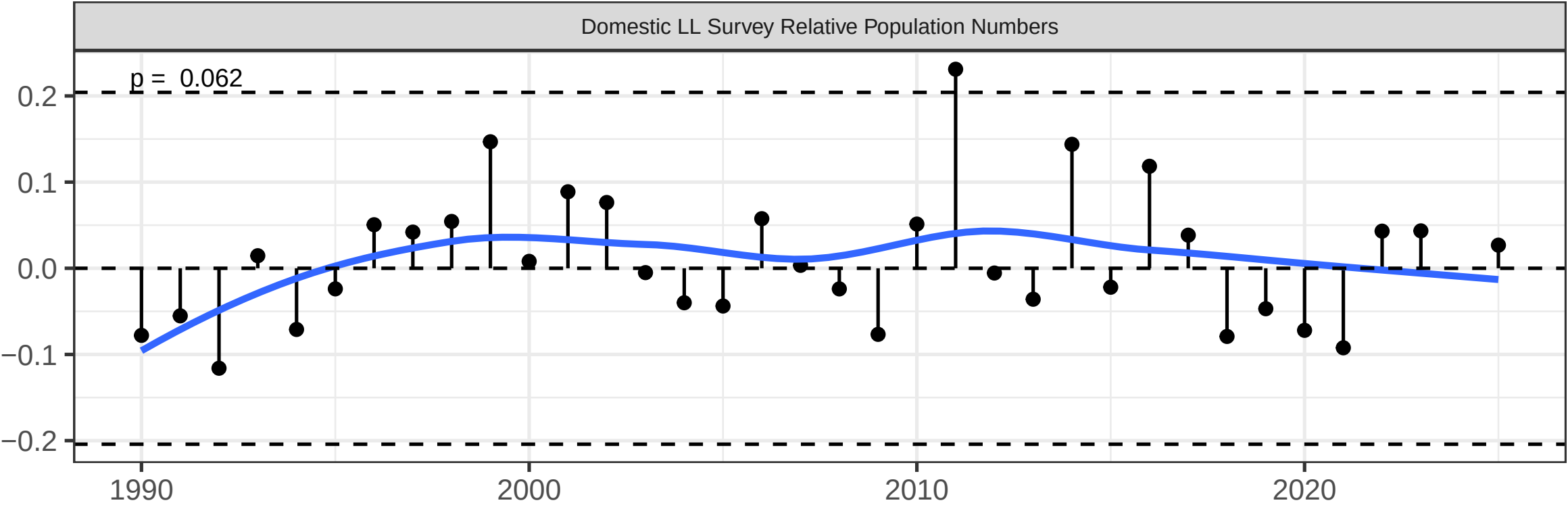
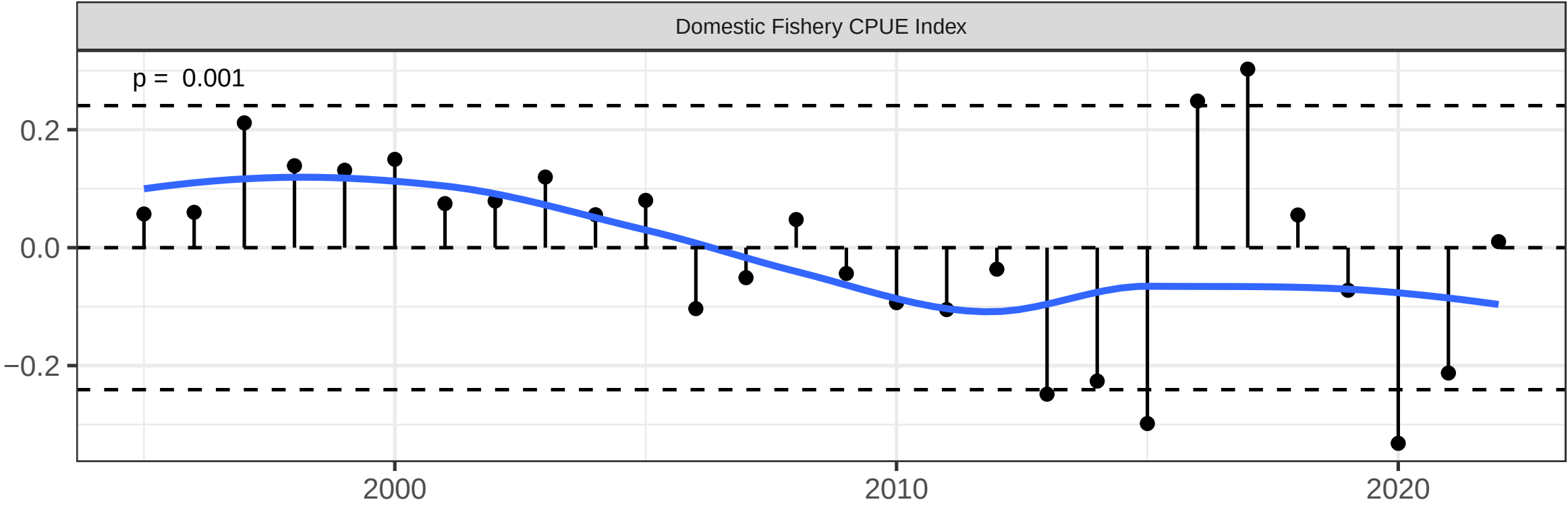




Index Fits

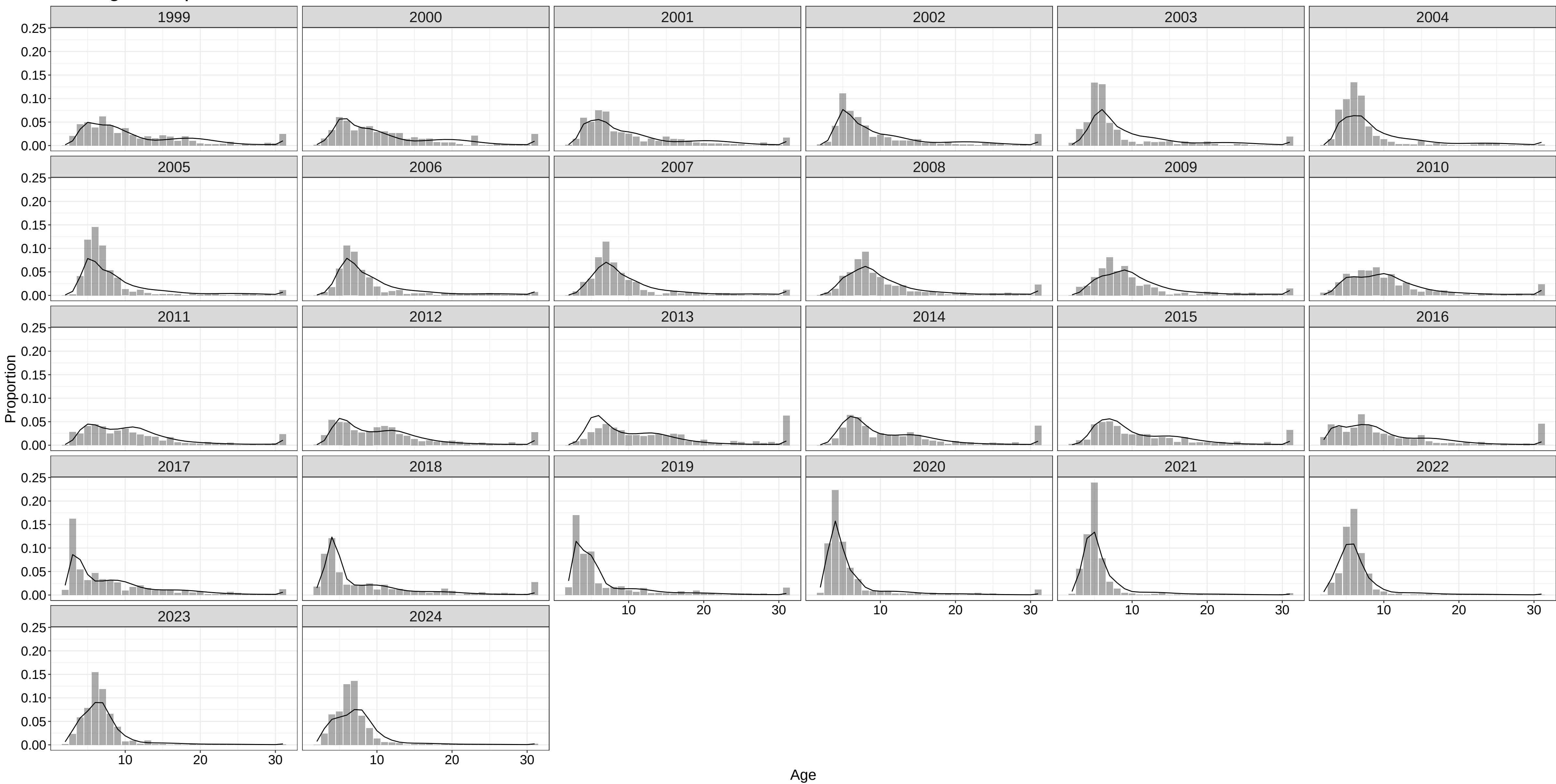


Index Residuals Runs Test

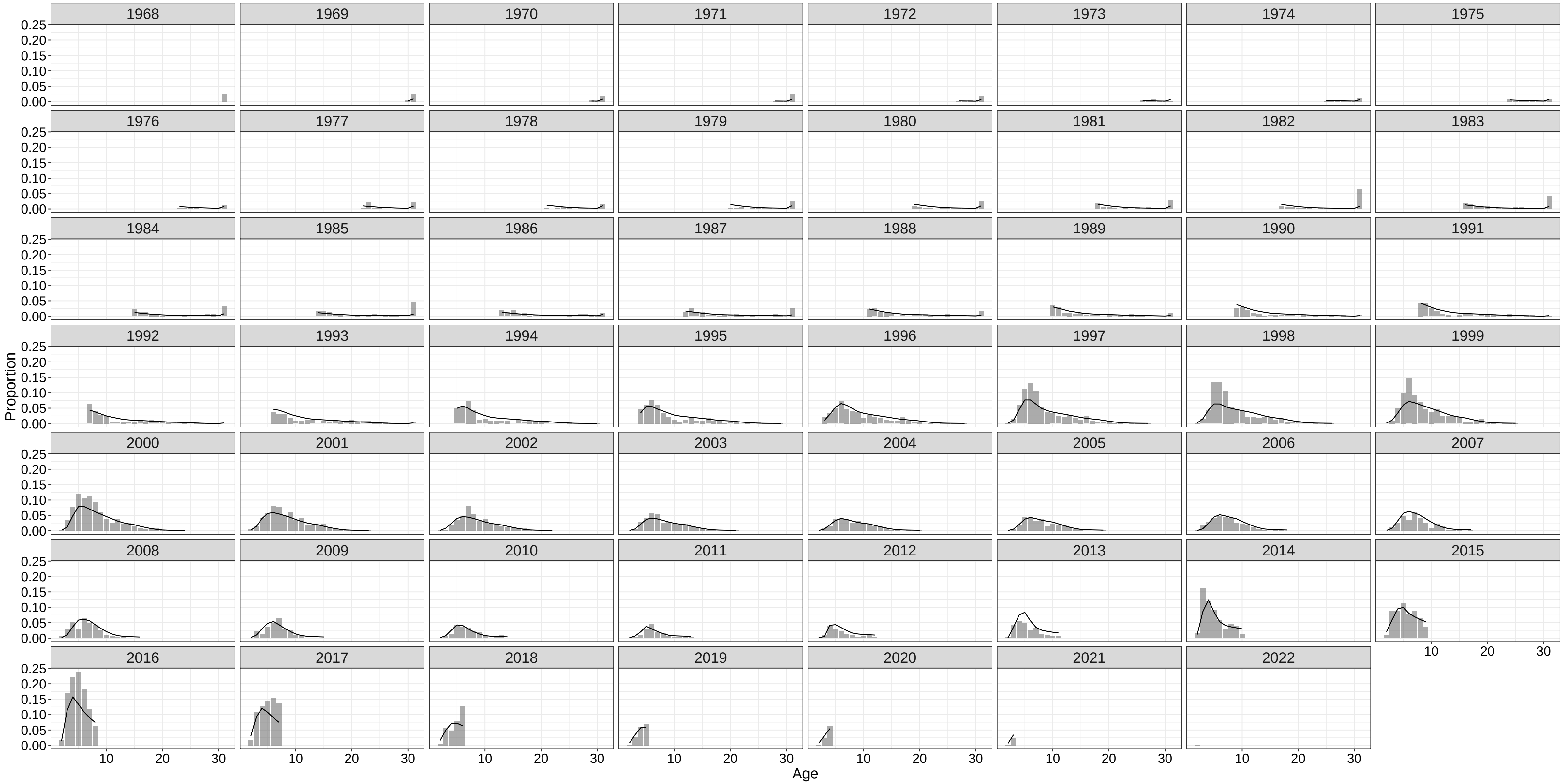


Year

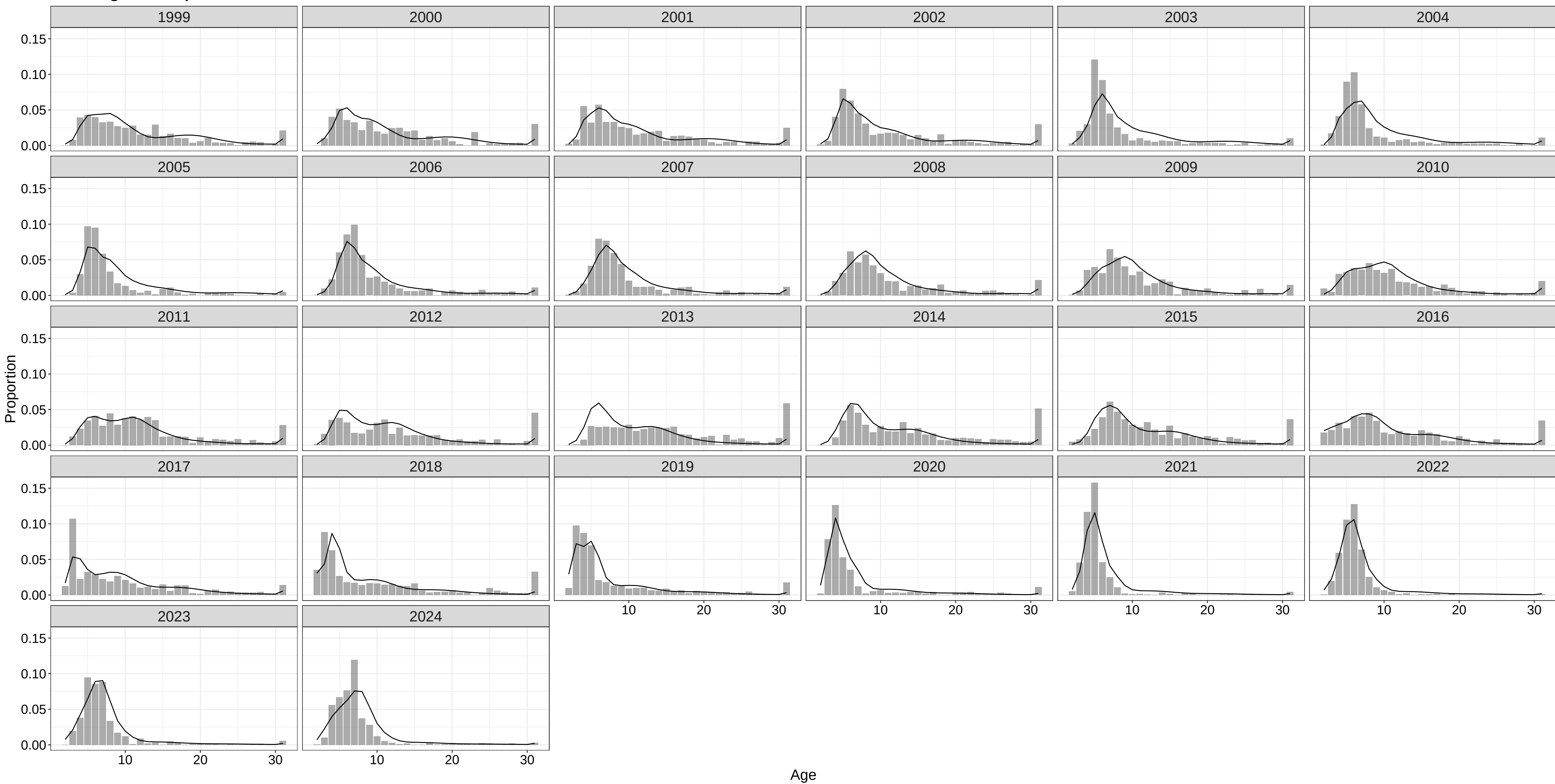
LLF Age Comps Female



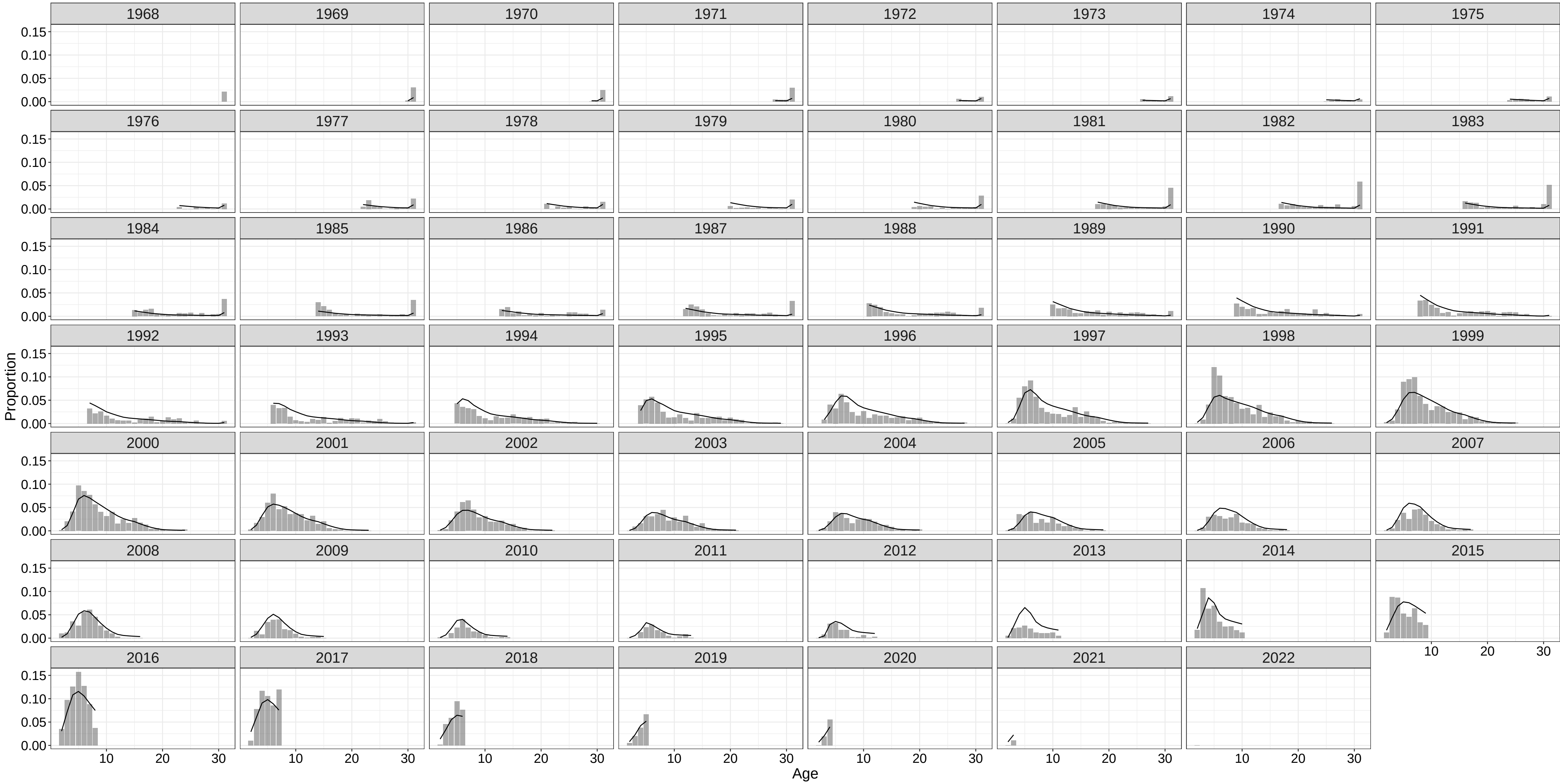
LLF Age Comps Cohort Fits Female



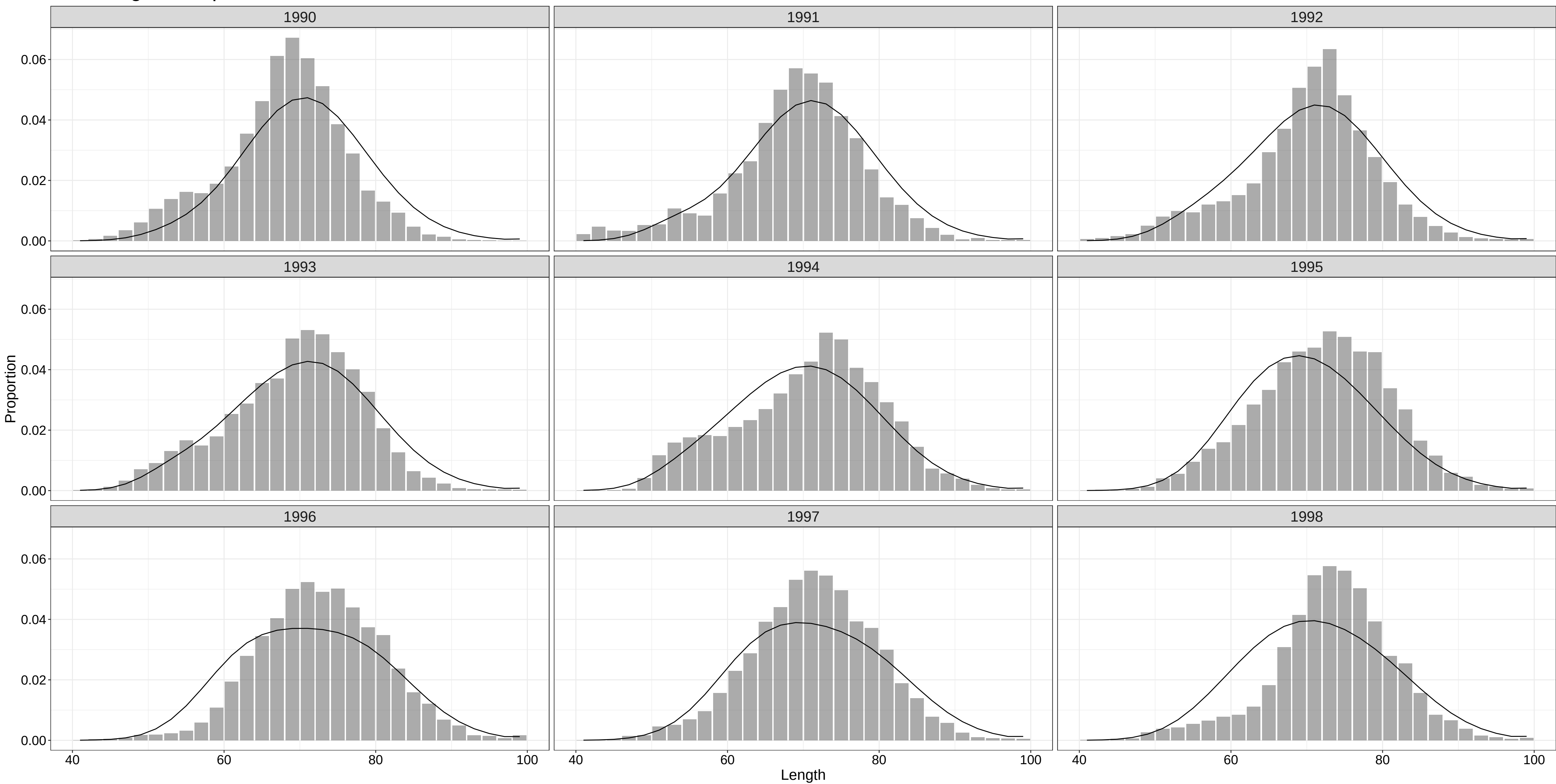
LLF Age Comps Male



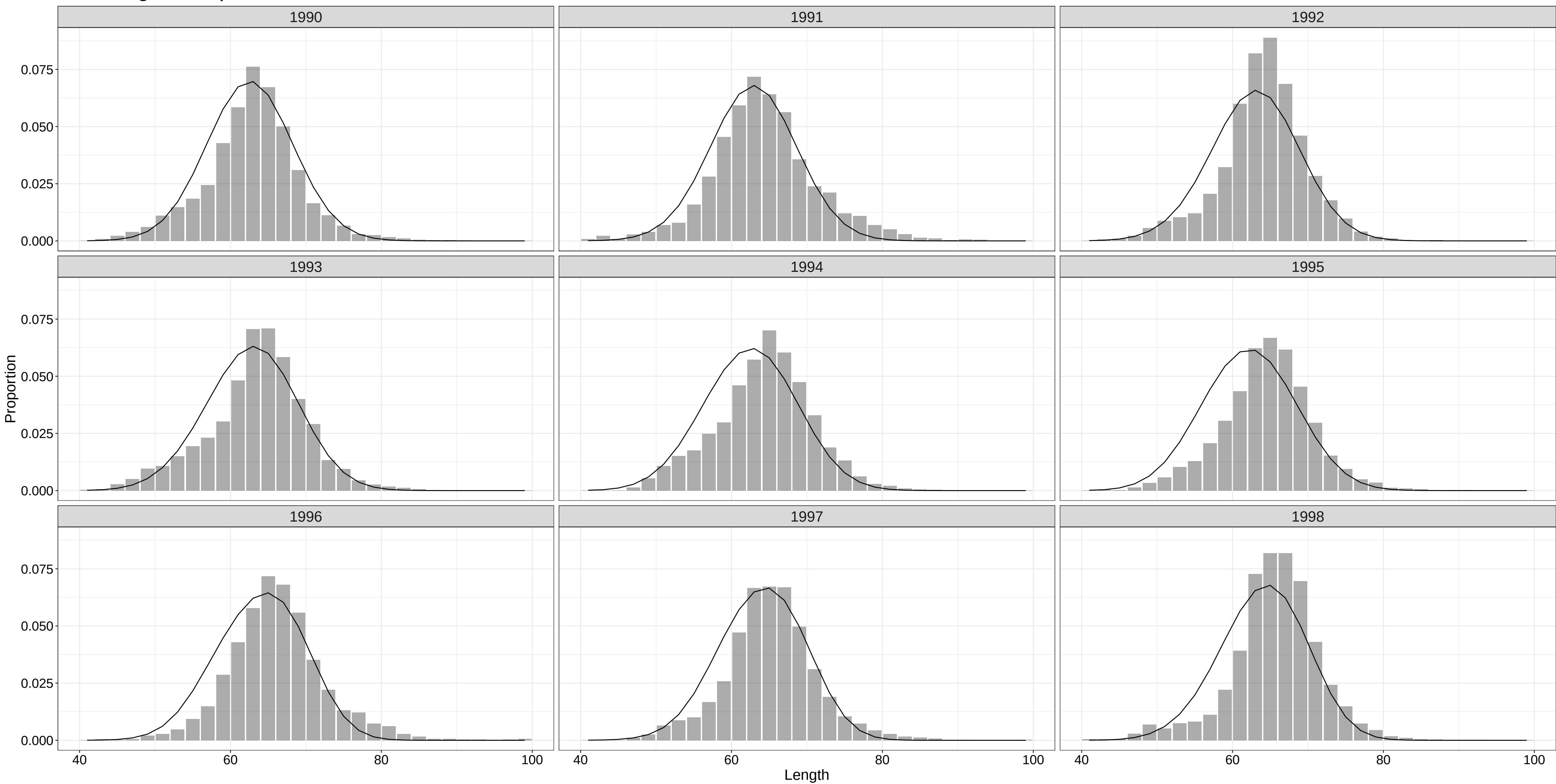
LLF Age Comps Cohort Fits Male



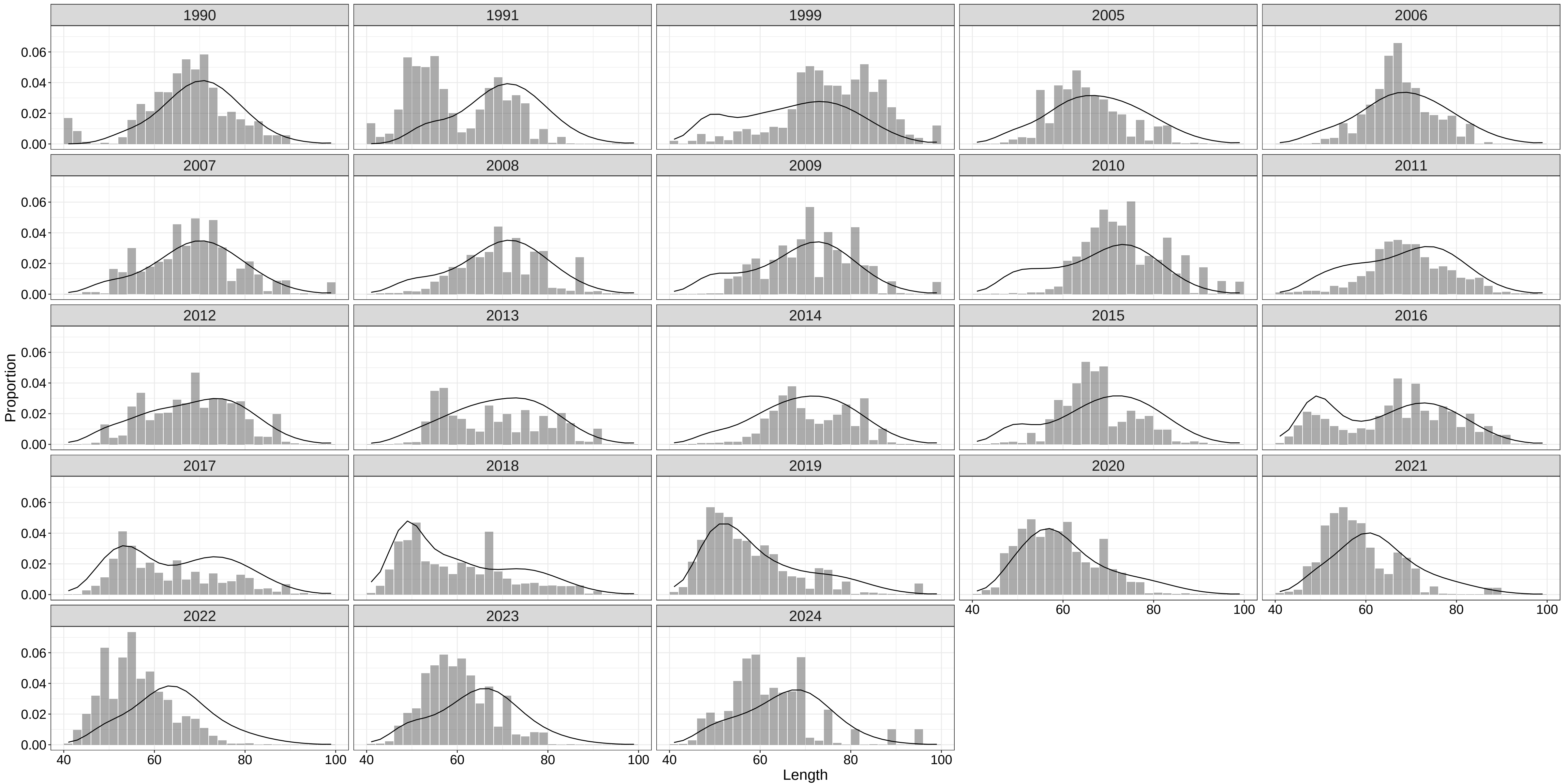
LLF Length Comps Female



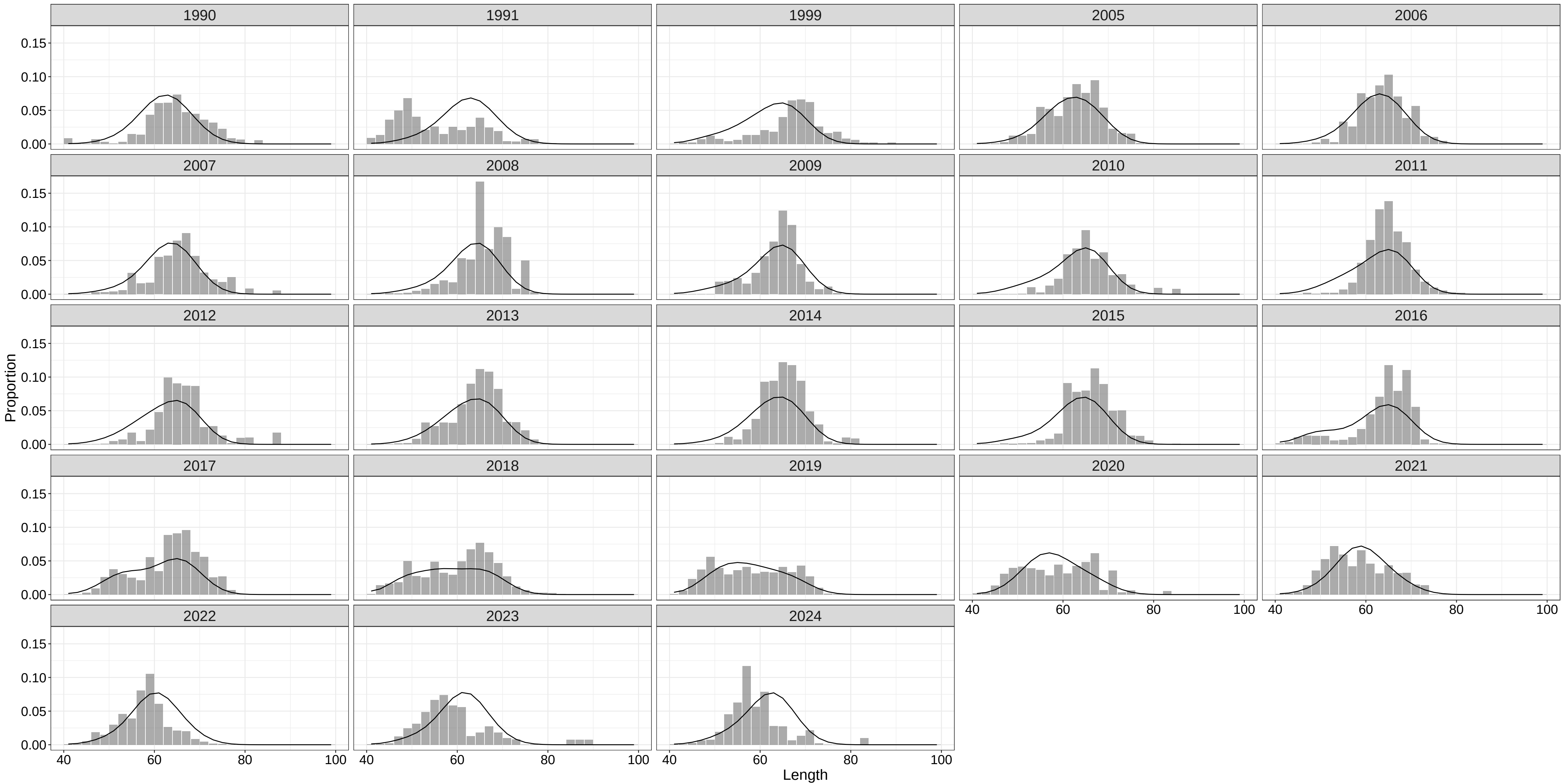
LLF Length Comps Male



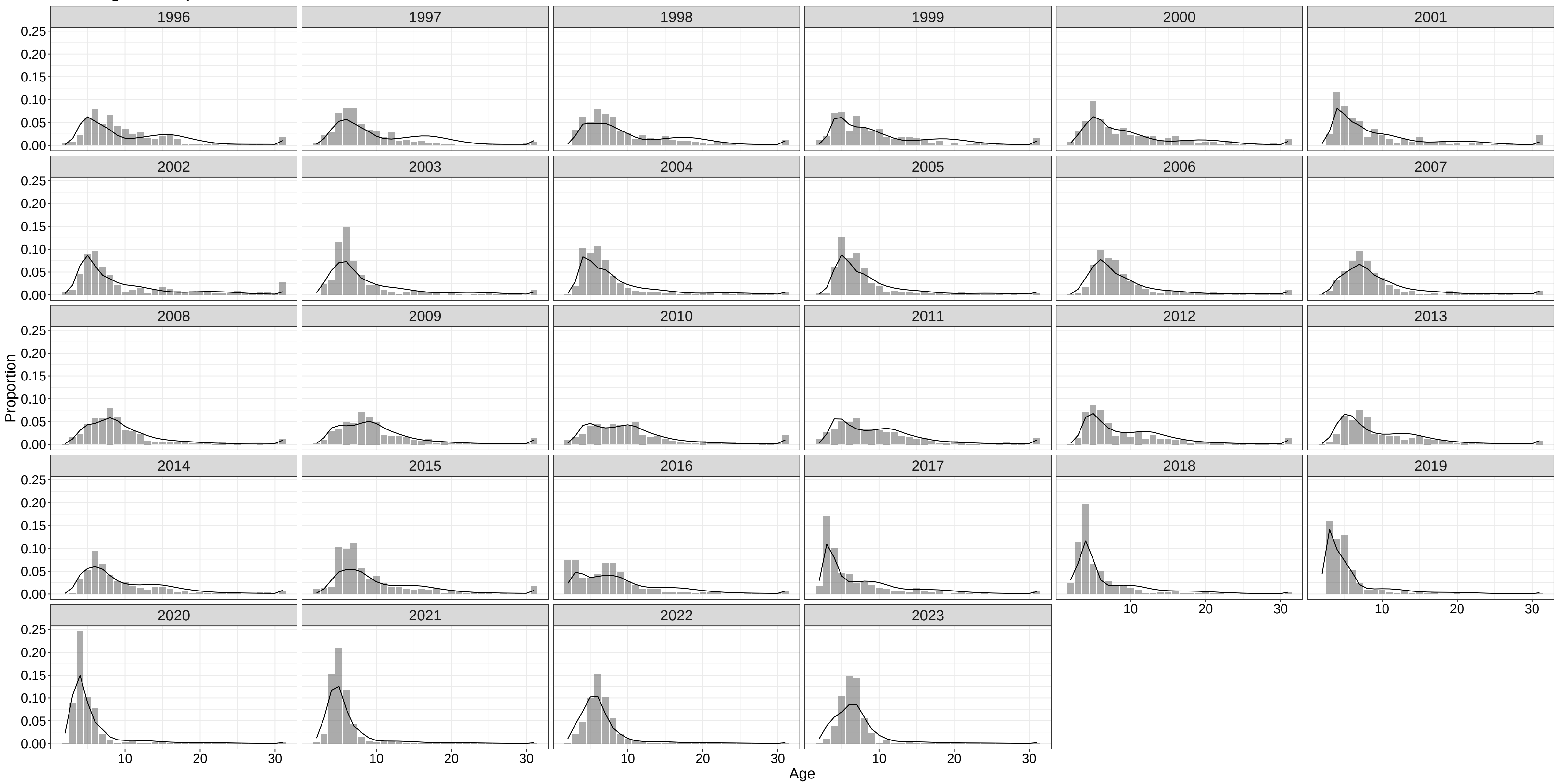
TF Length Comps Female



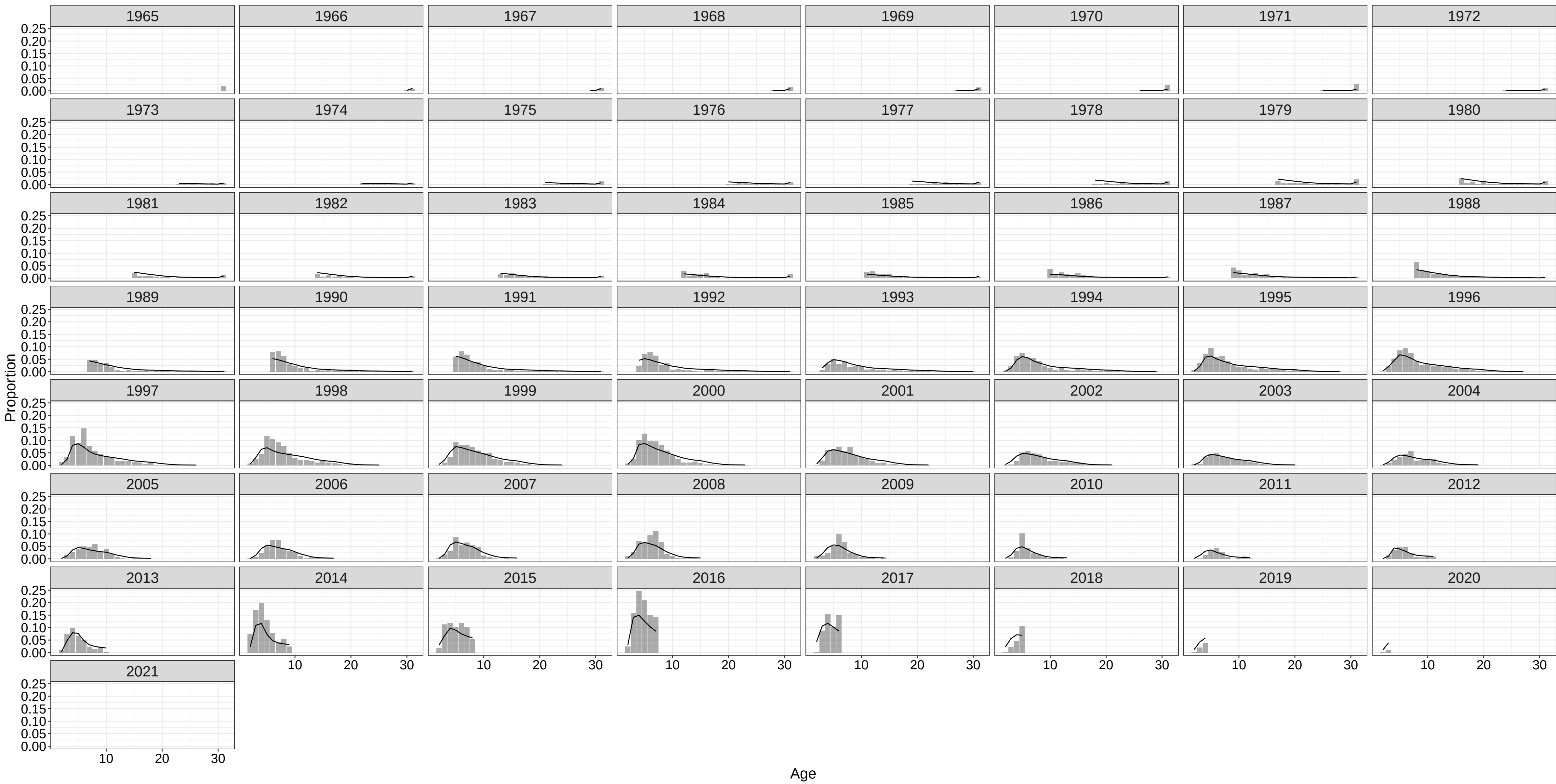
TF Length Comps Male



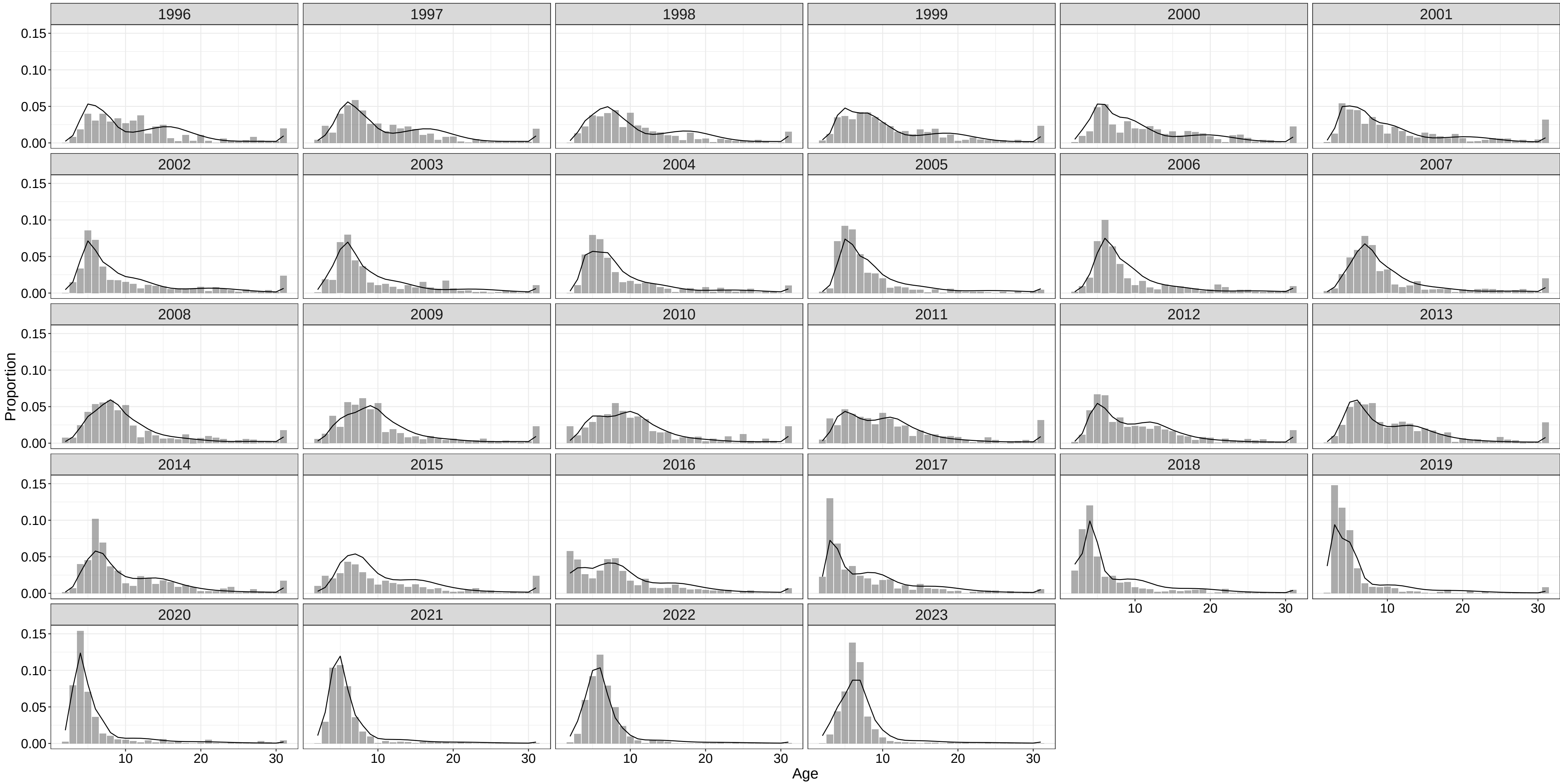
LLS Age Comps Female



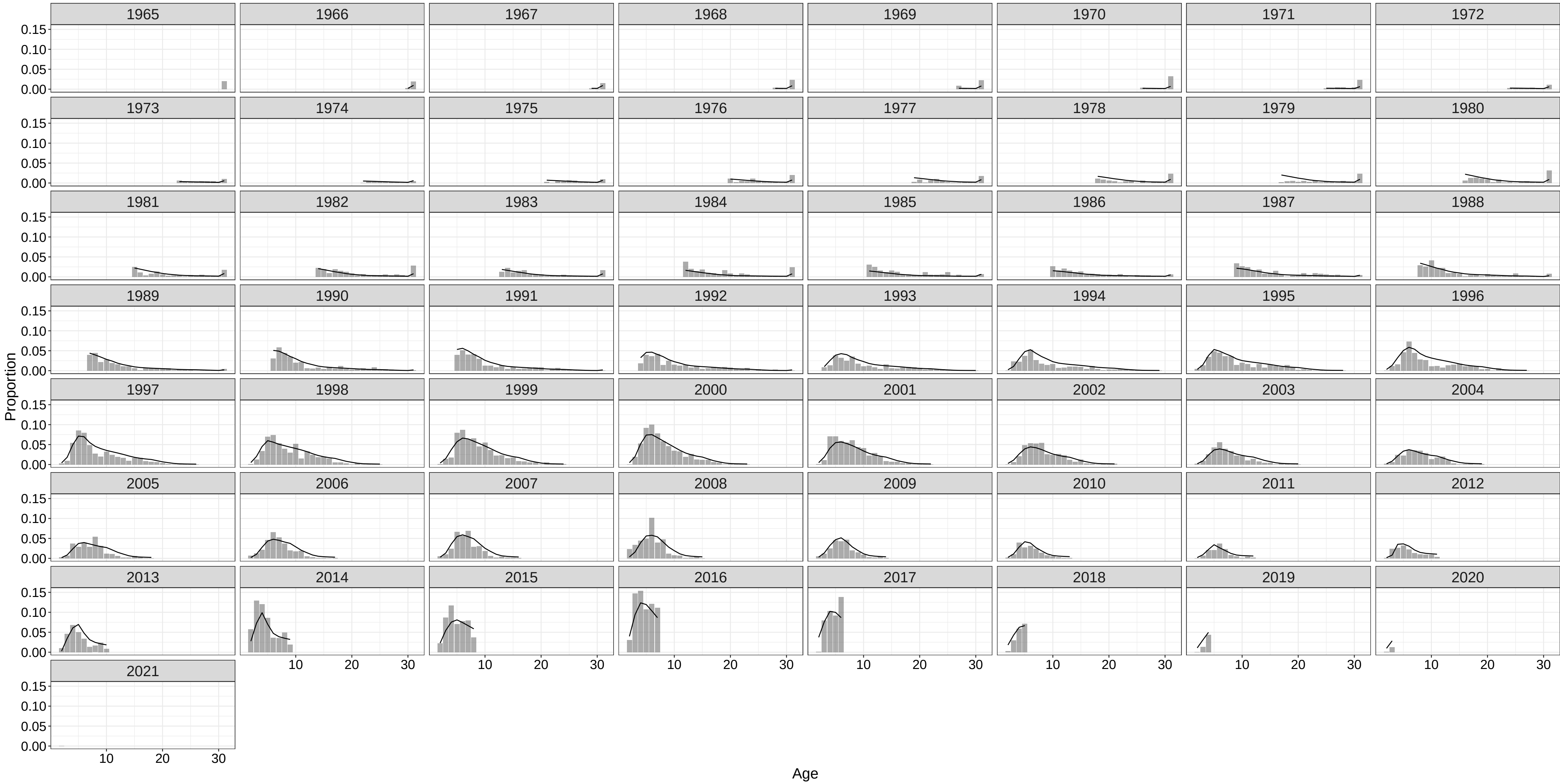
LLS Age Comps Cohort Fit Female



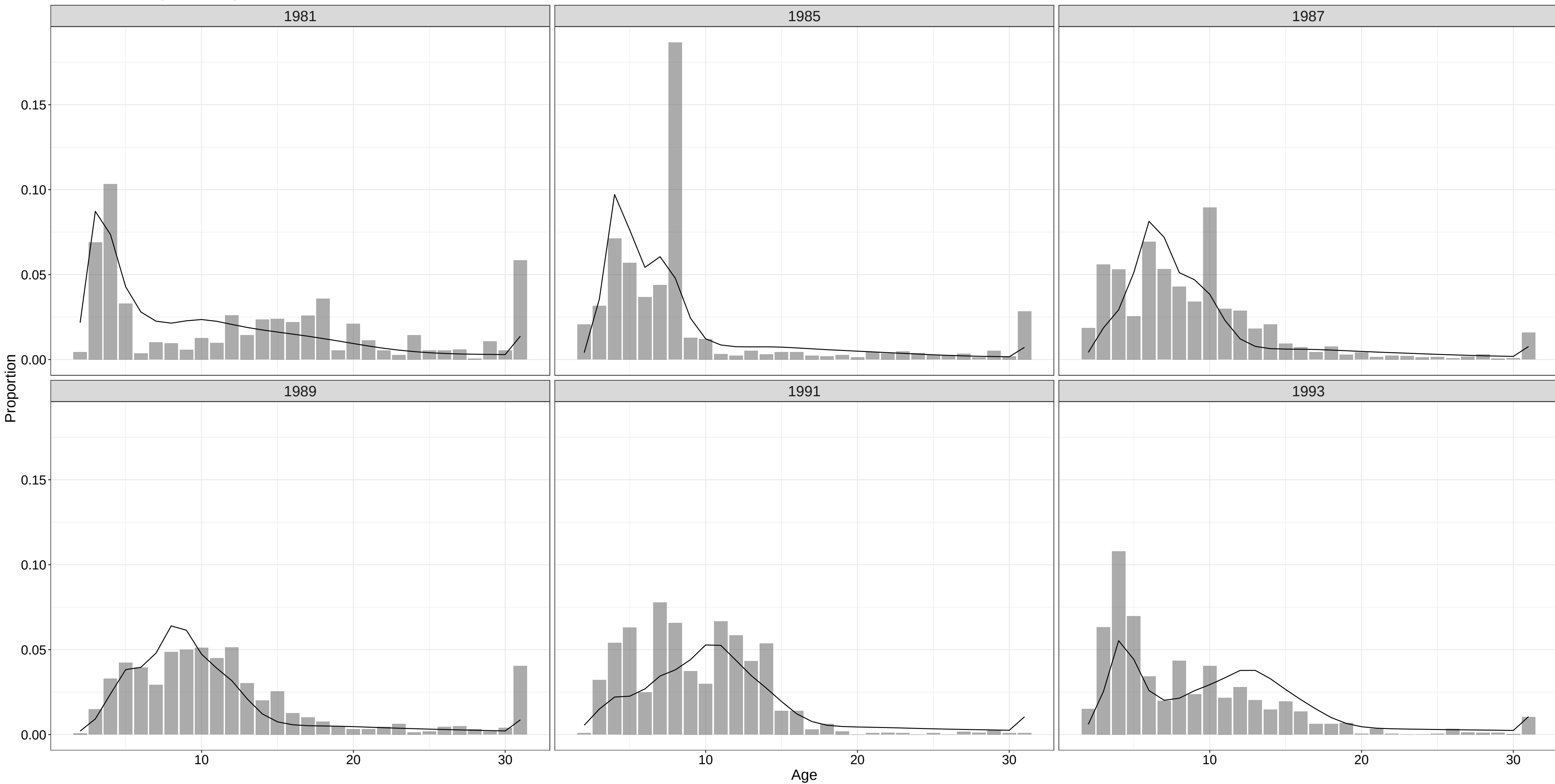
LLS Age Comps Male



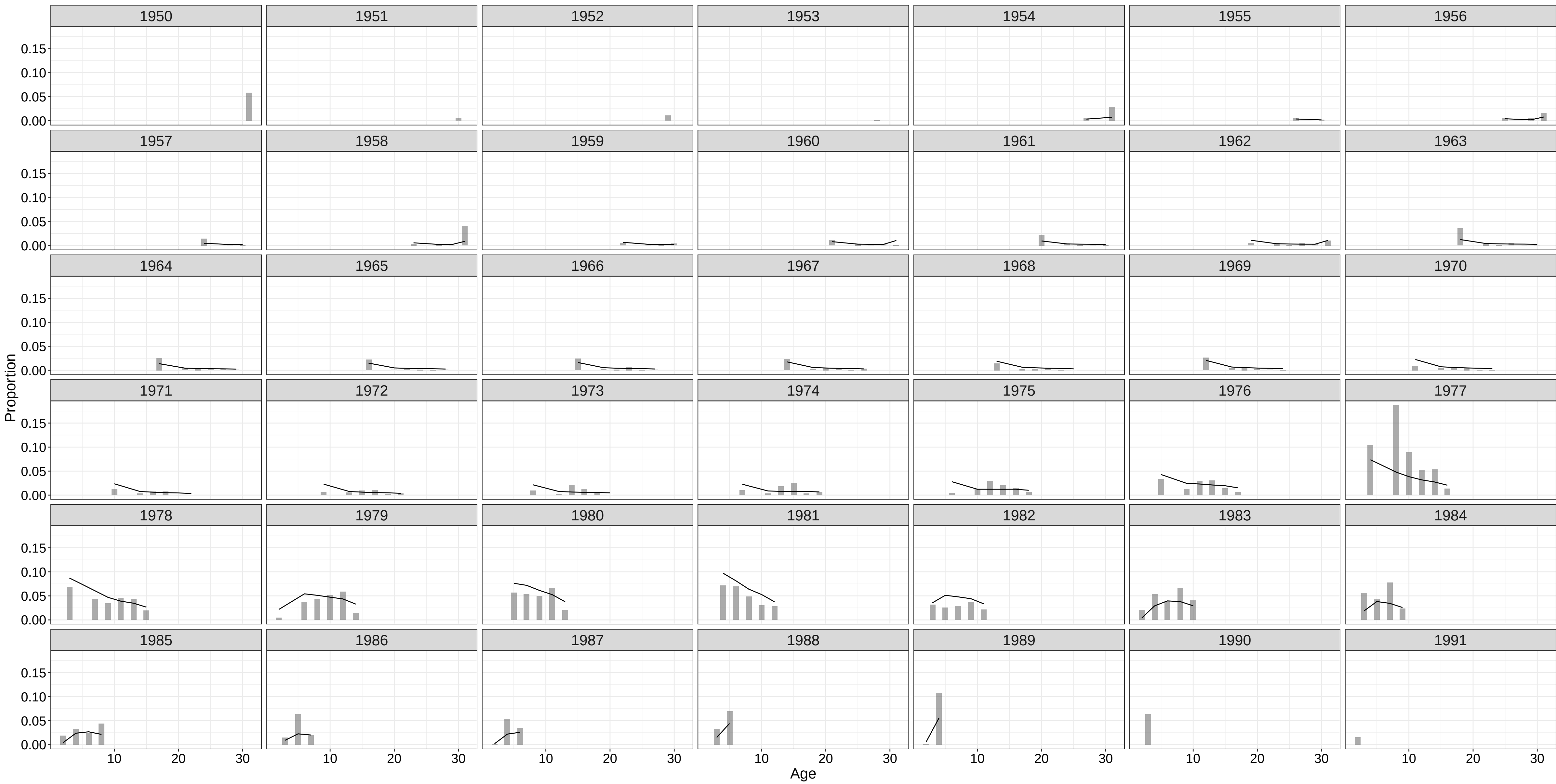
LLS Age Comps Cohort Fit Male



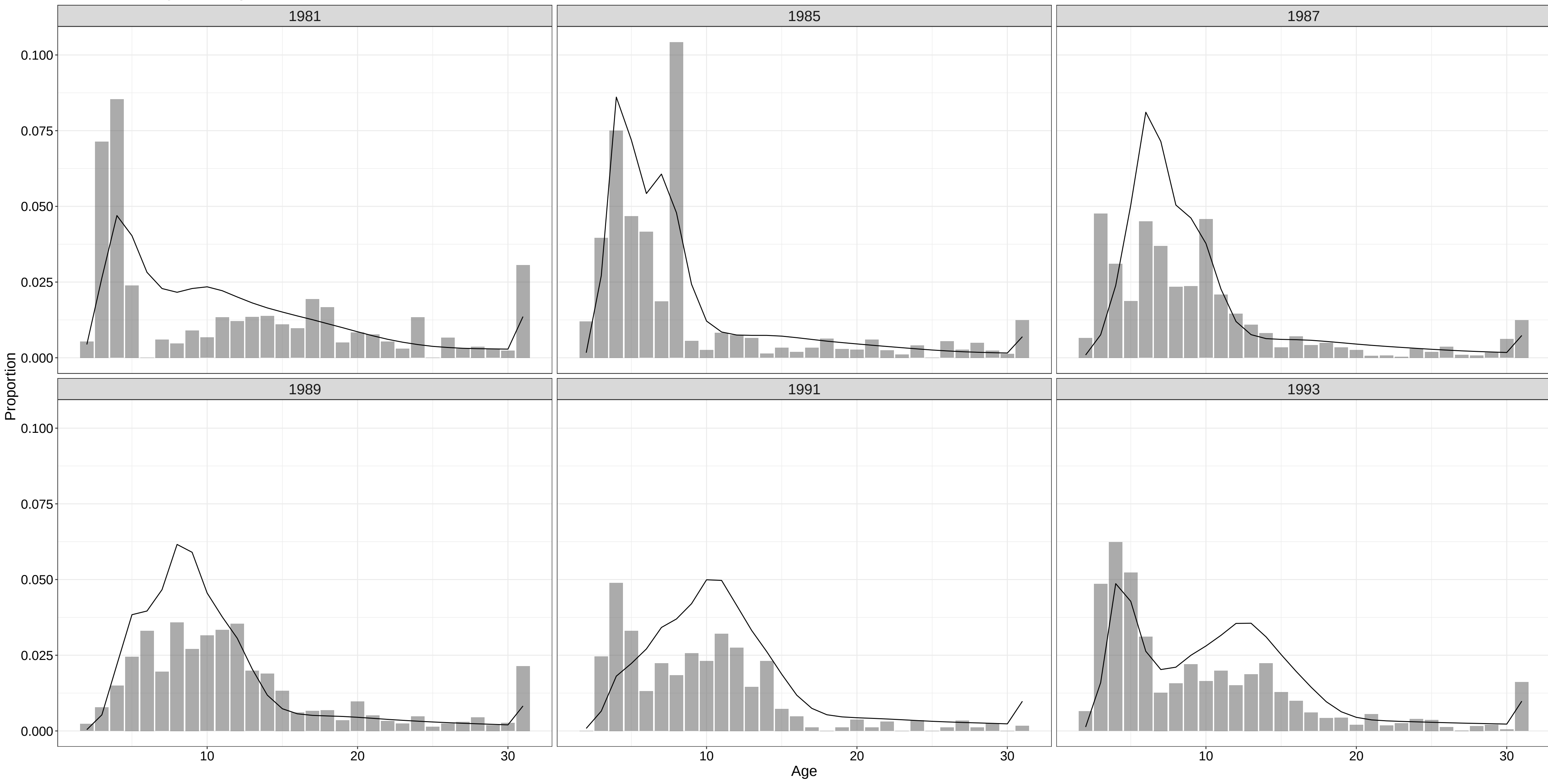
JPN LLS Age Comps Female



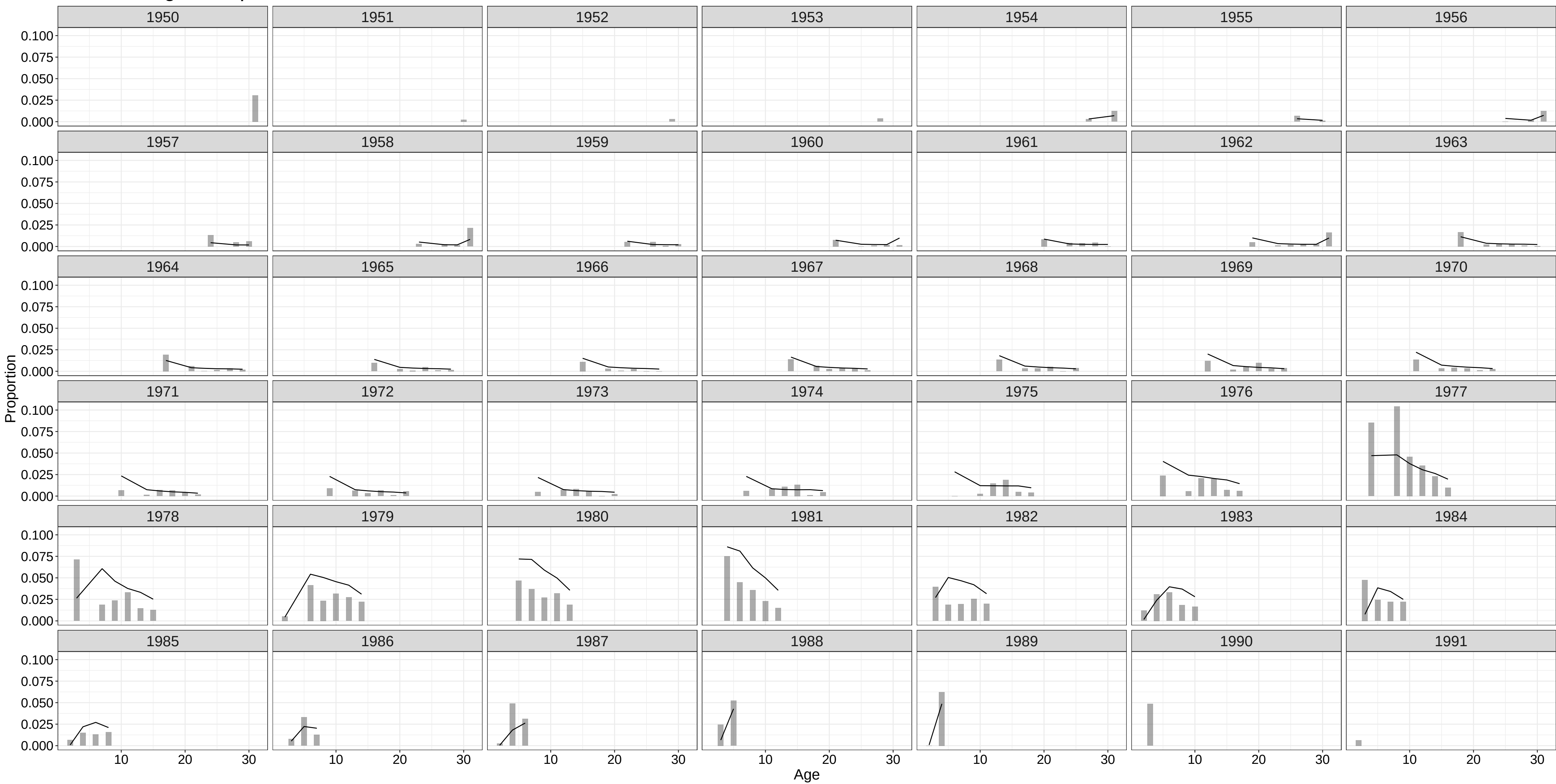
JPN LLS Age Comps Cohort Fit Female



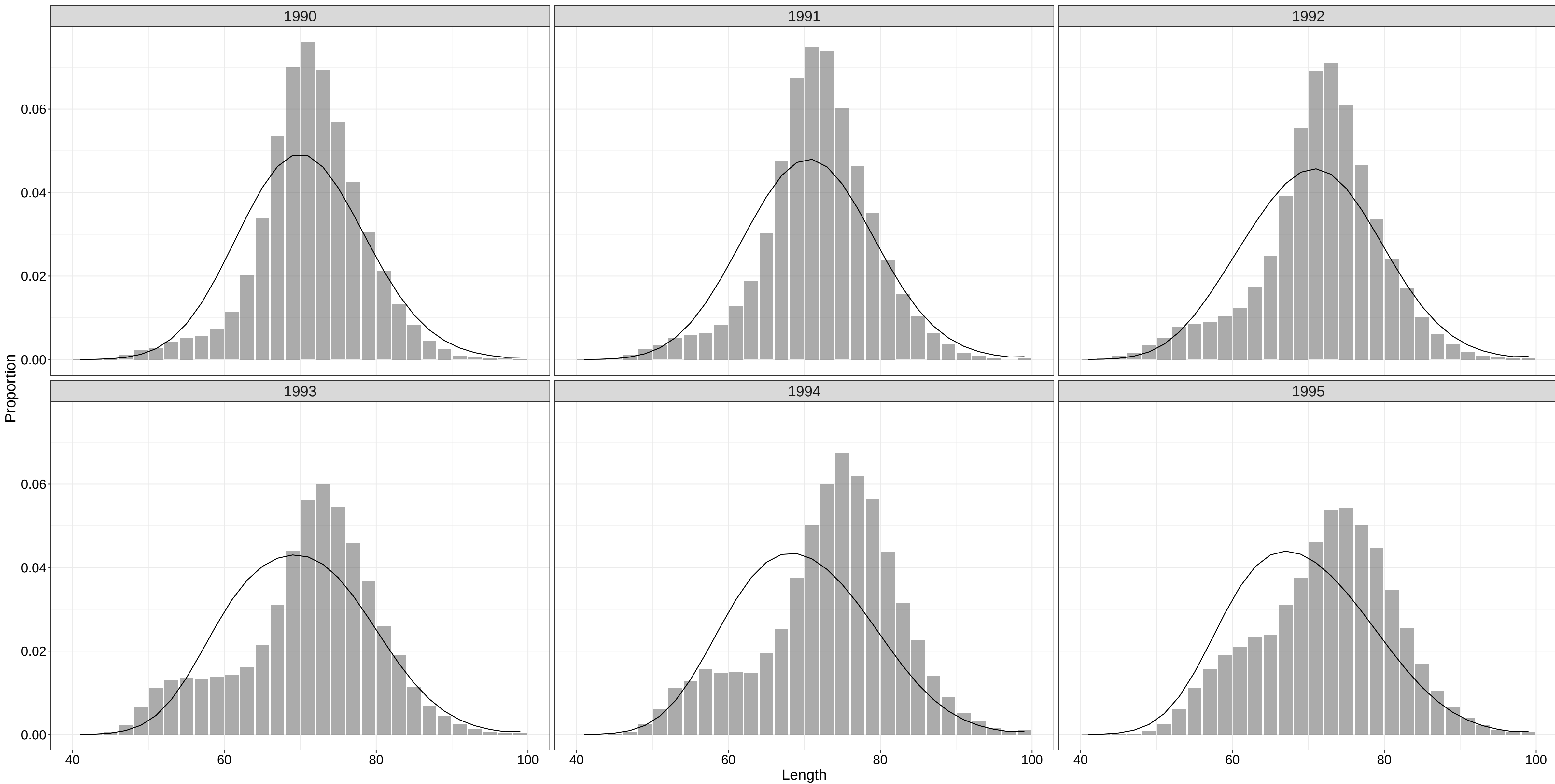
JPN LLS Age Comps Male



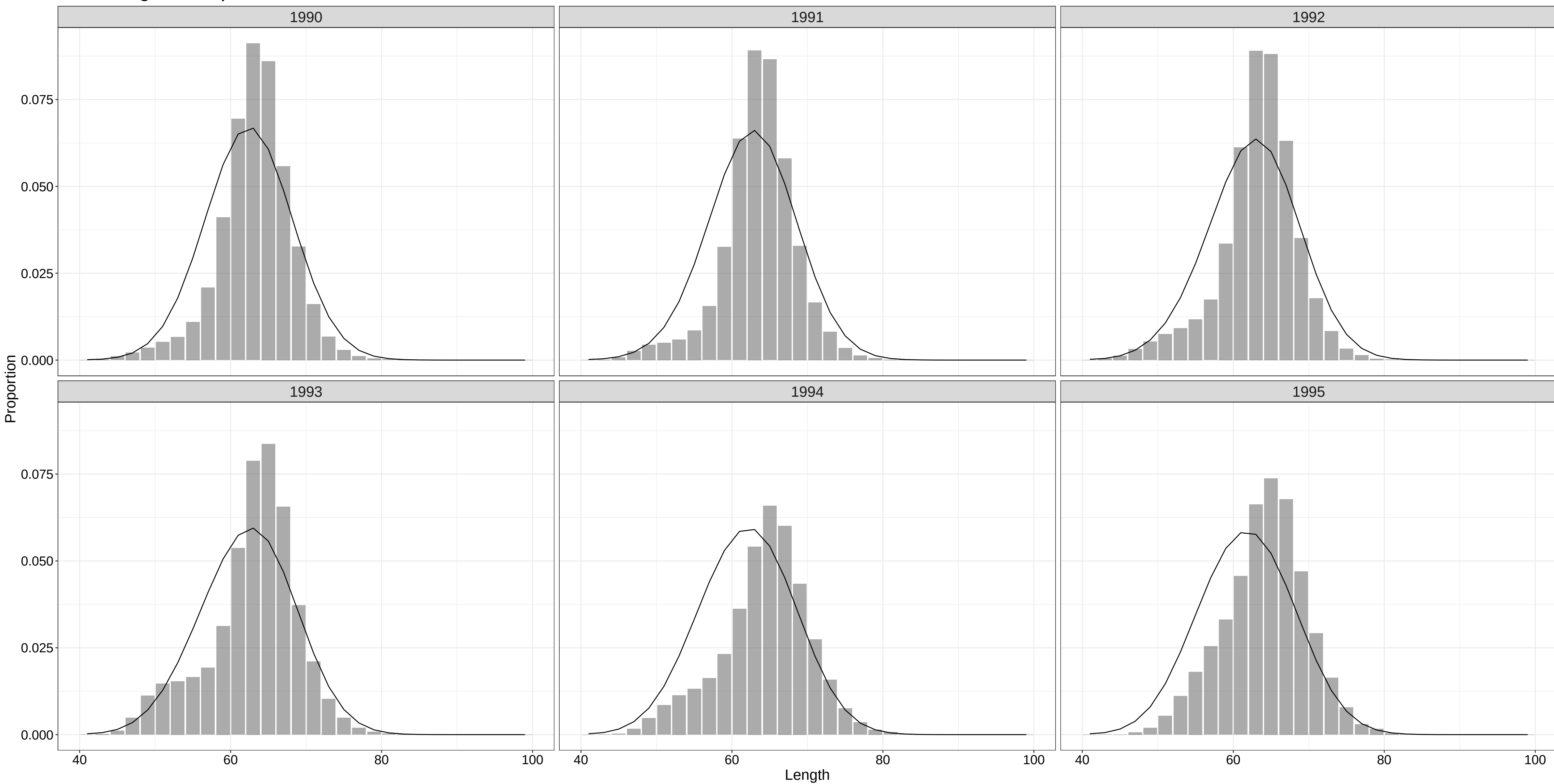
JPN LLS Age Comps Cohort Fit Male



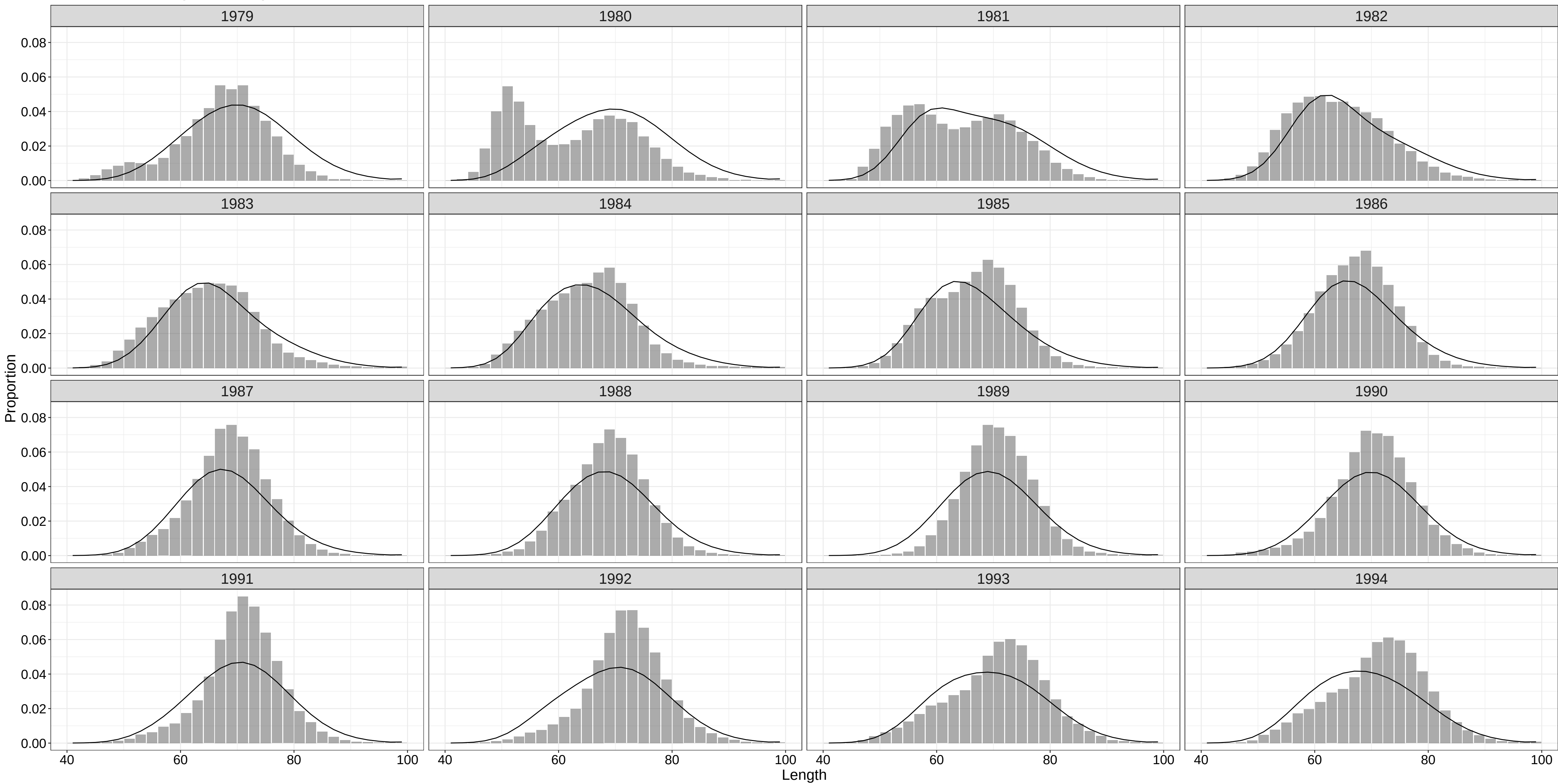
LLS Length Comps Female



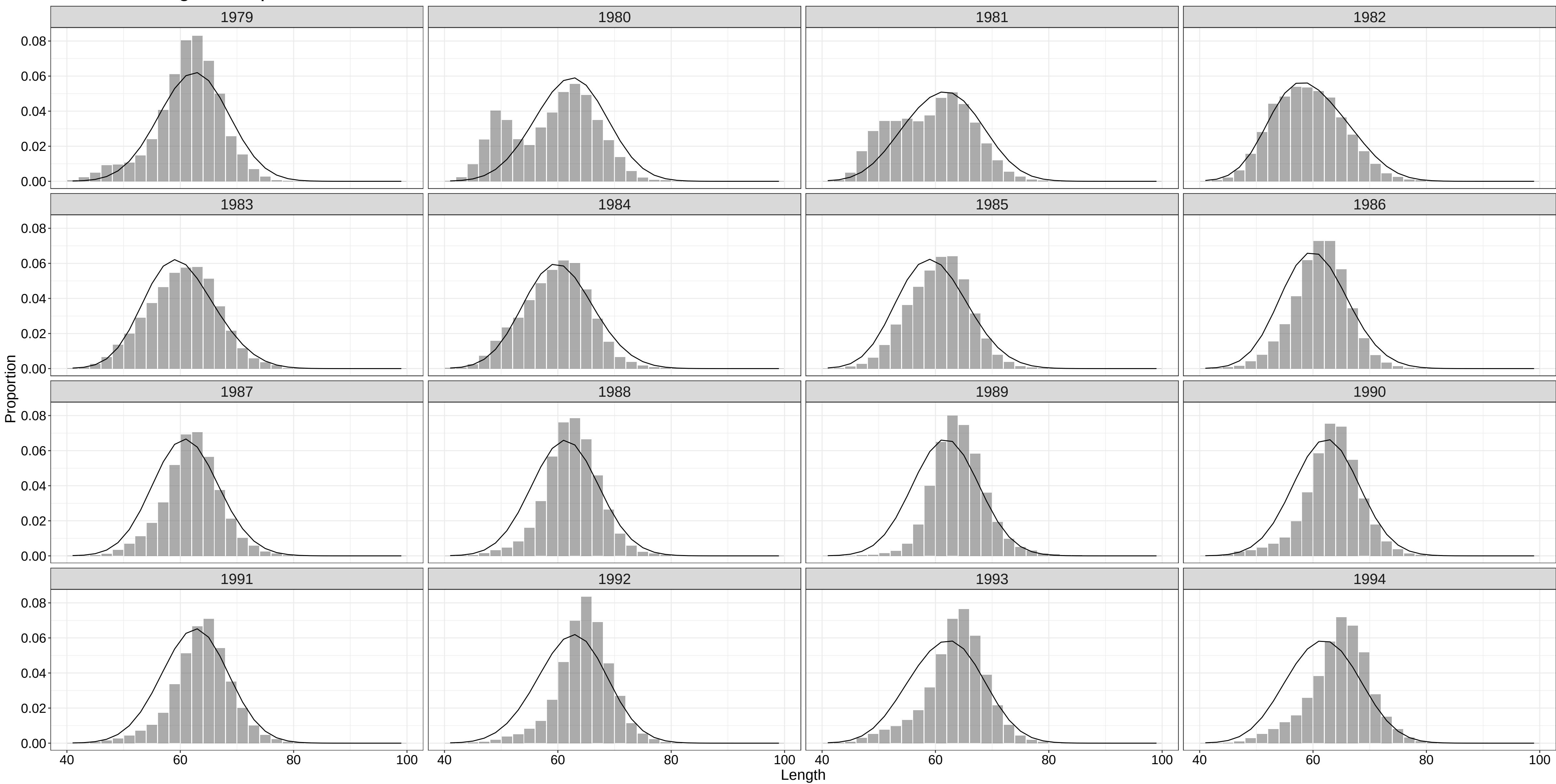
LLS Length Comps Male



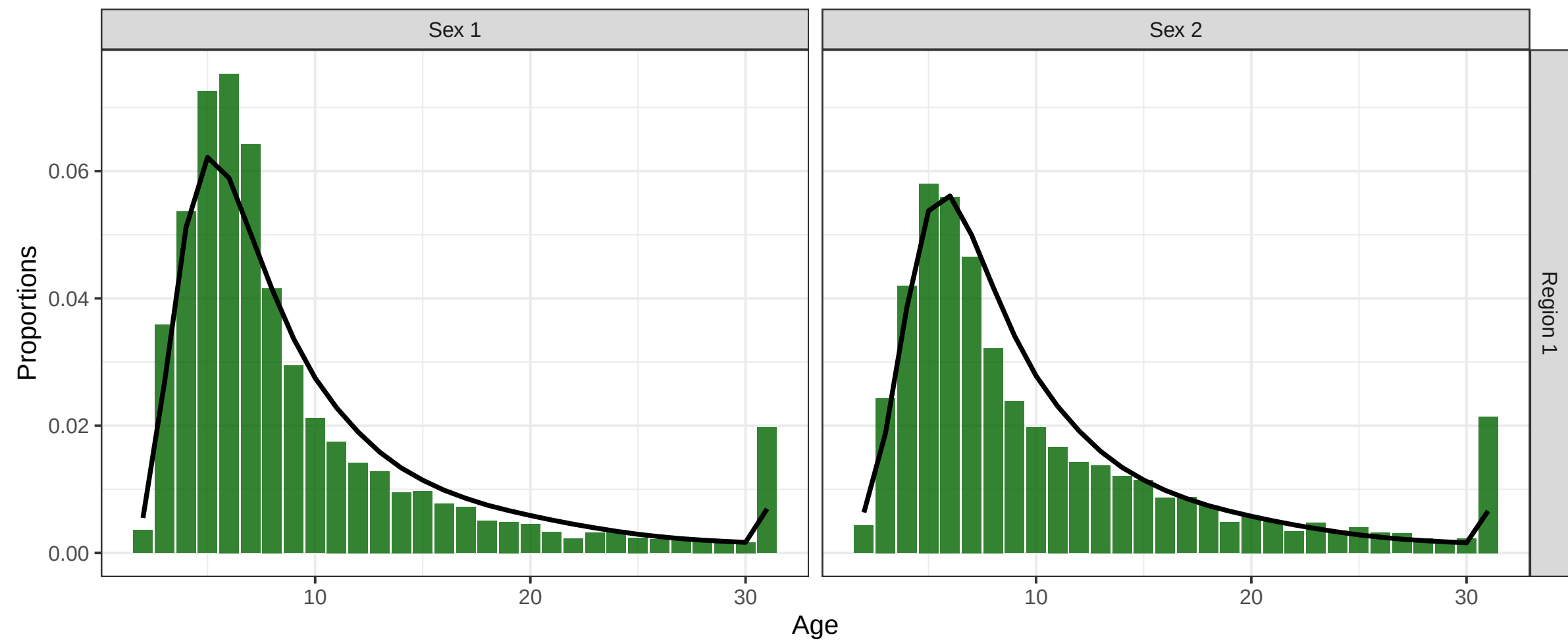
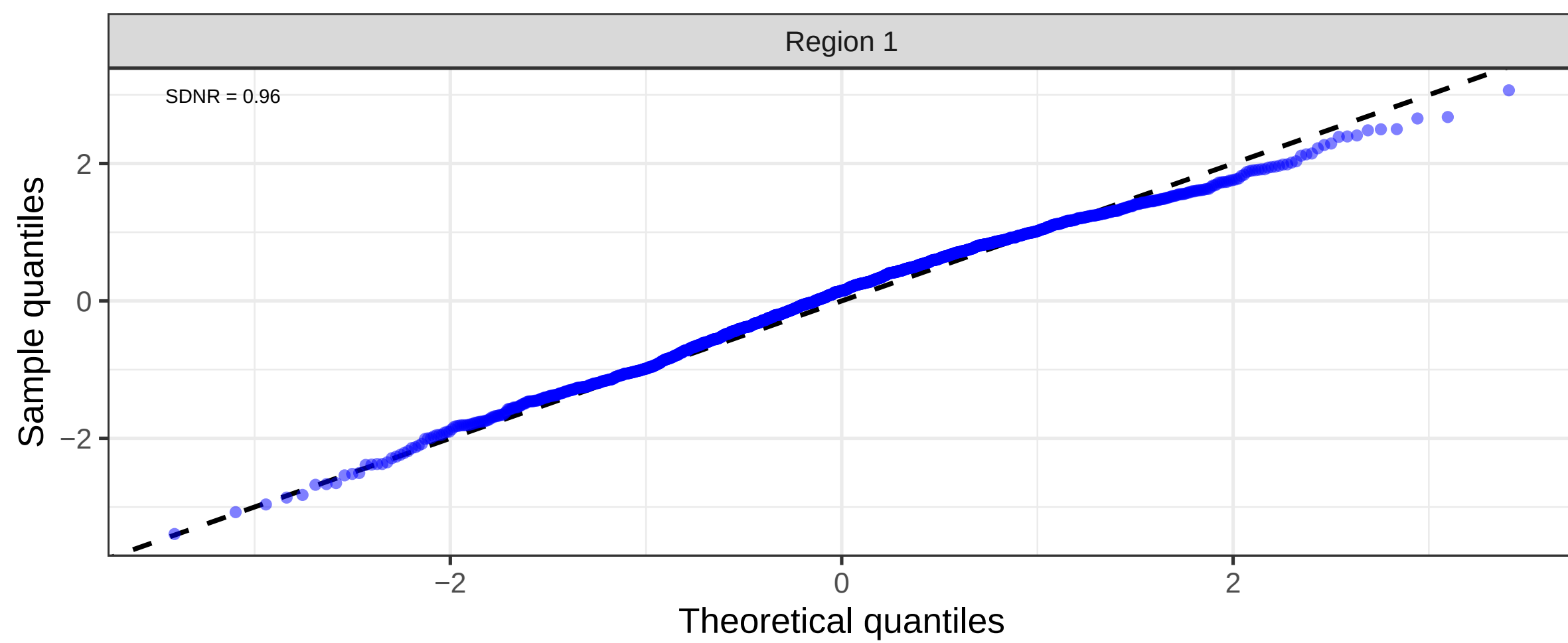
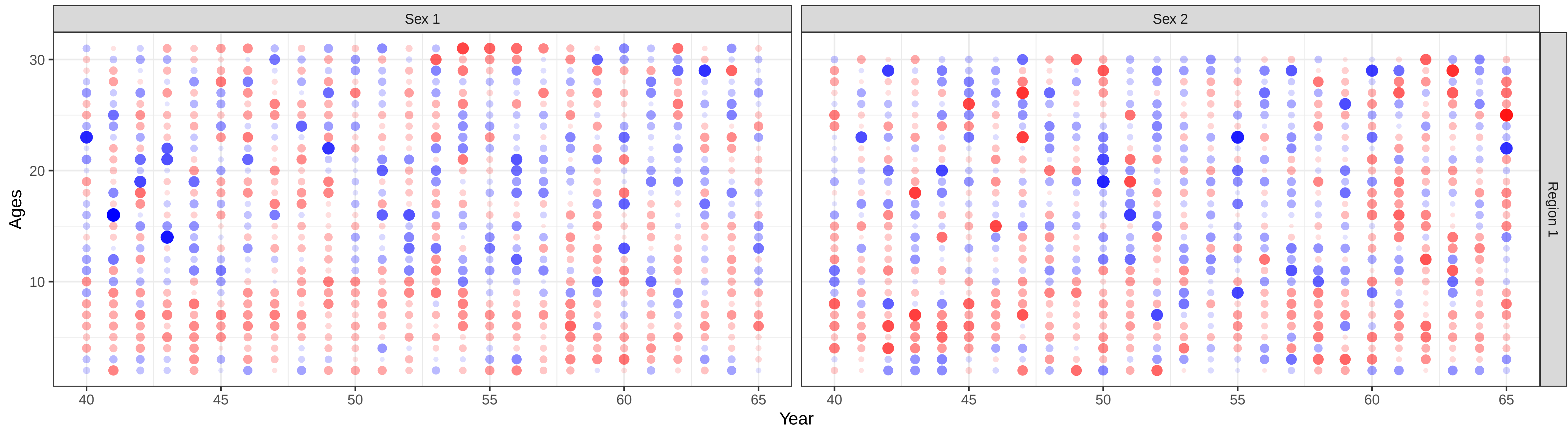
JPN LLS Length Comps Female



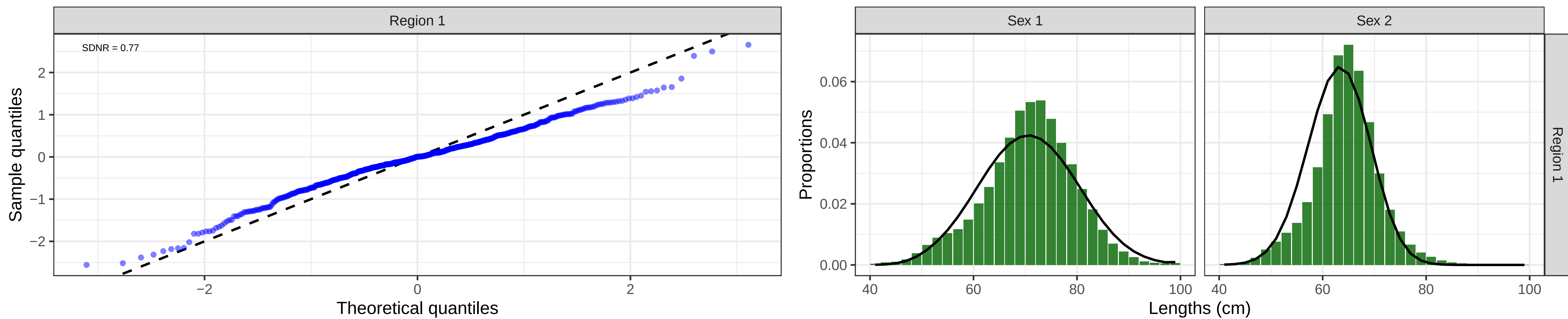
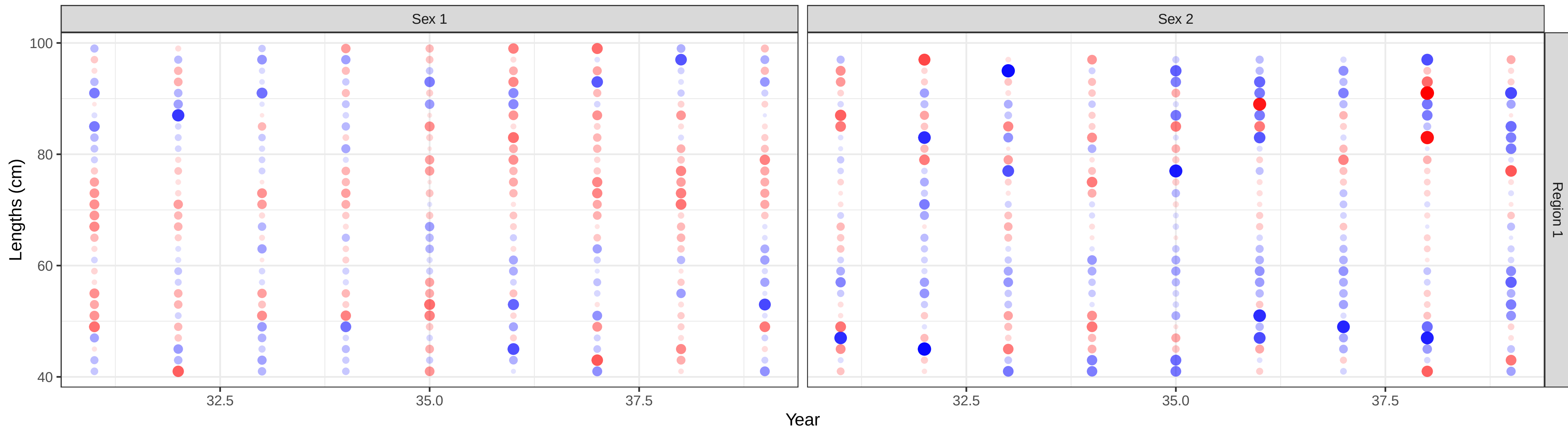
JPN LLS Length Comps Male

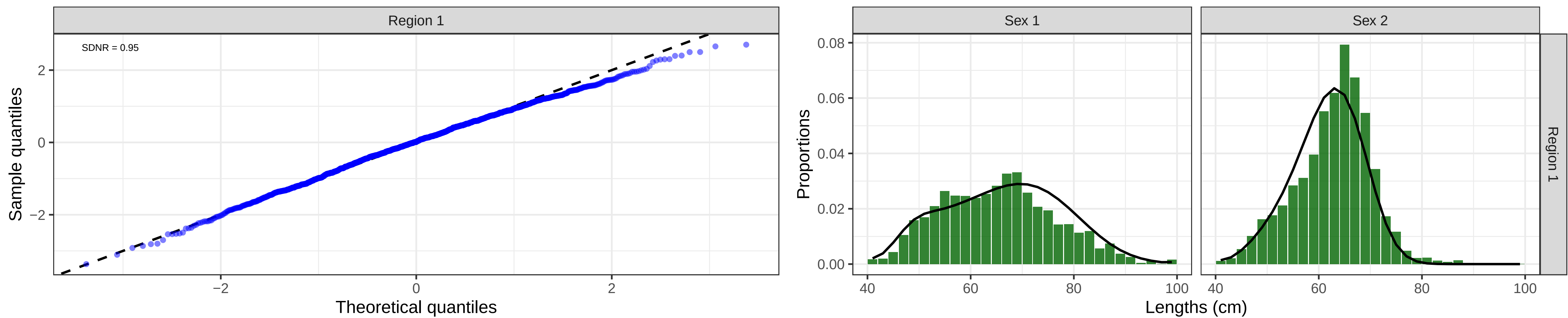
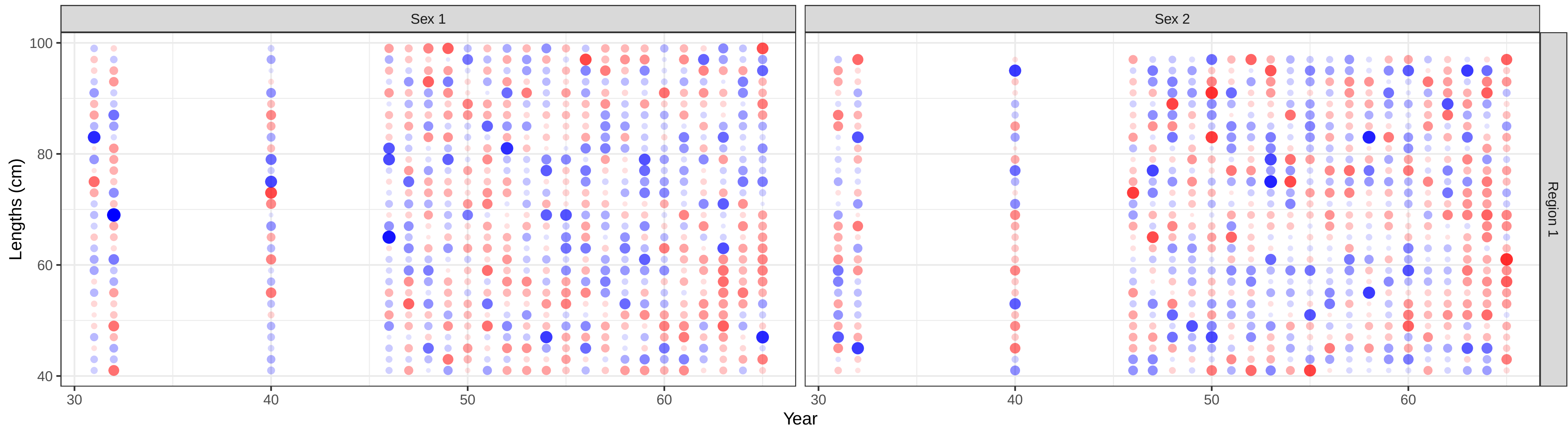


abs(Resid) 1 2 3 Sign ● Neg ● Pos

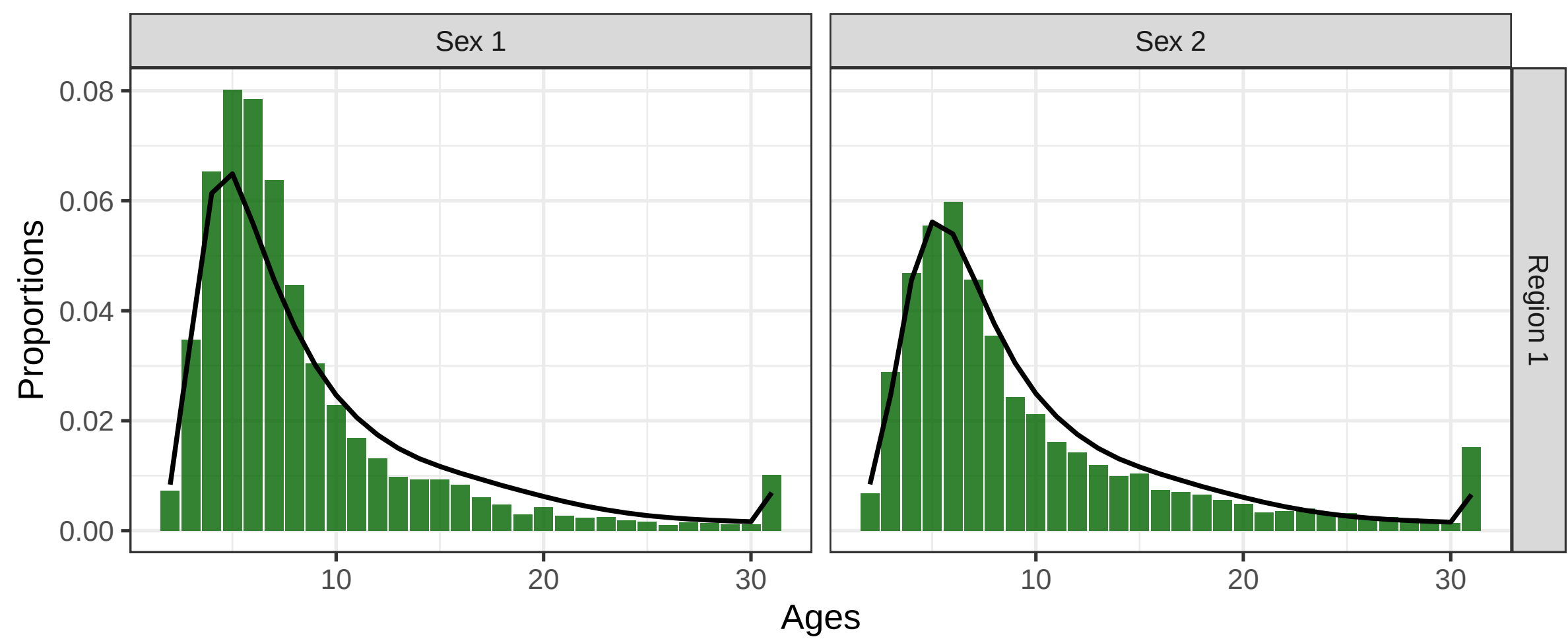
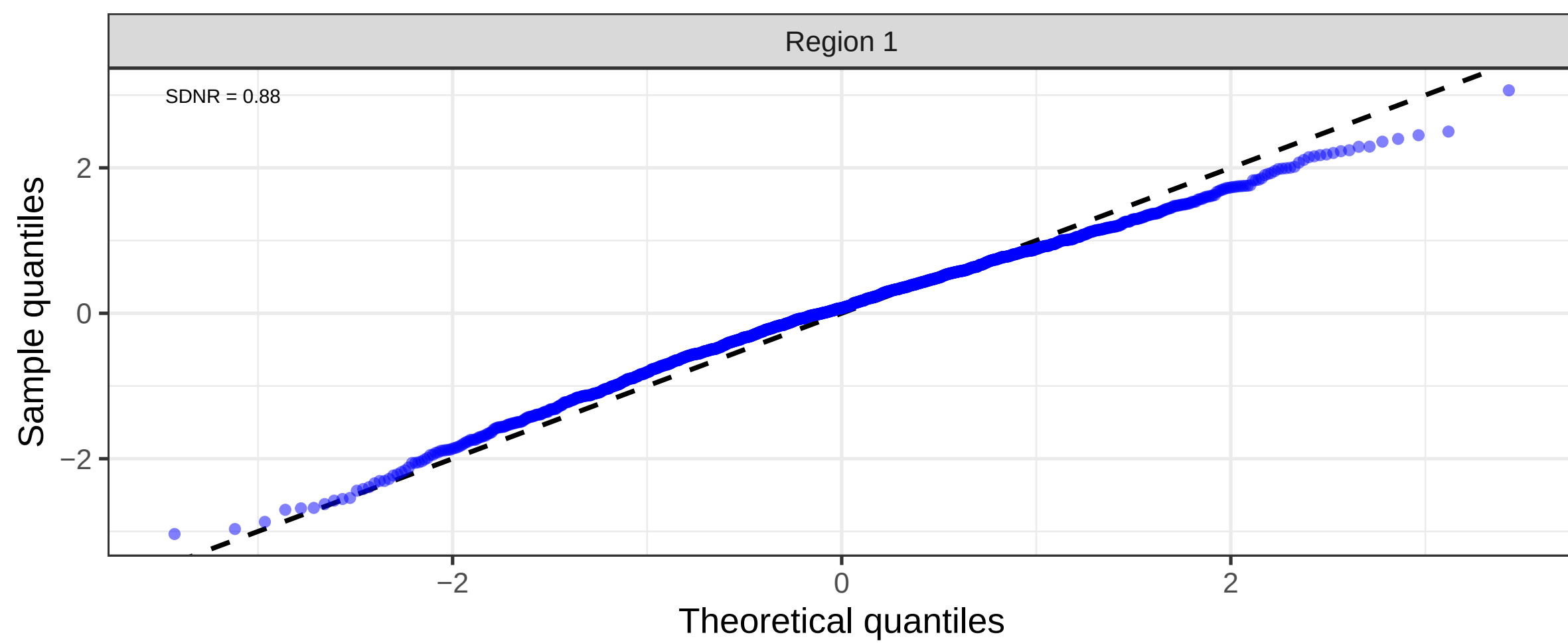
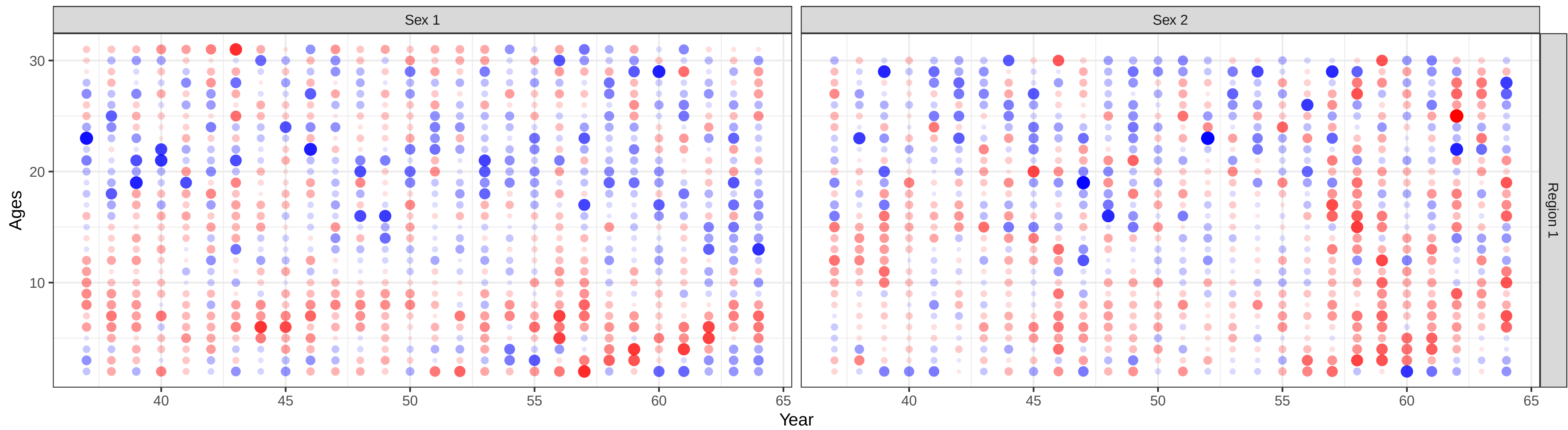


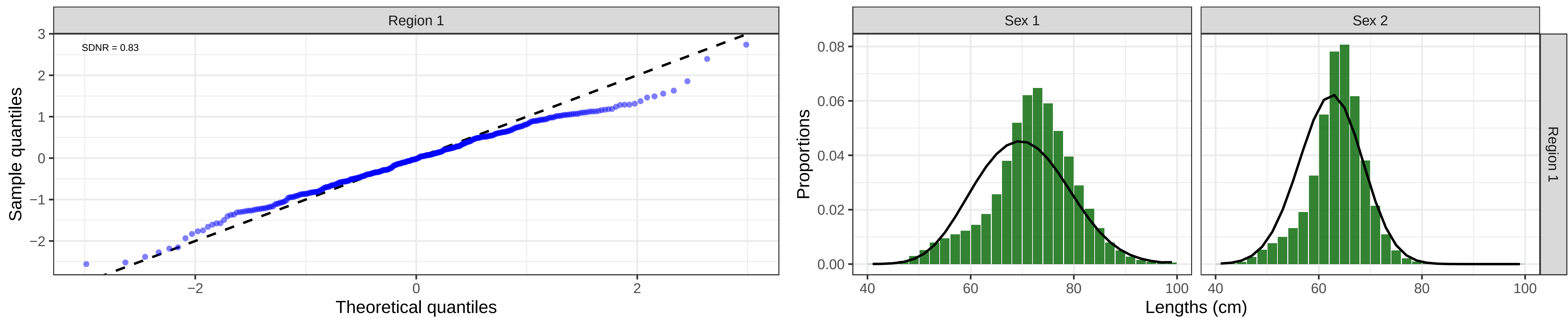
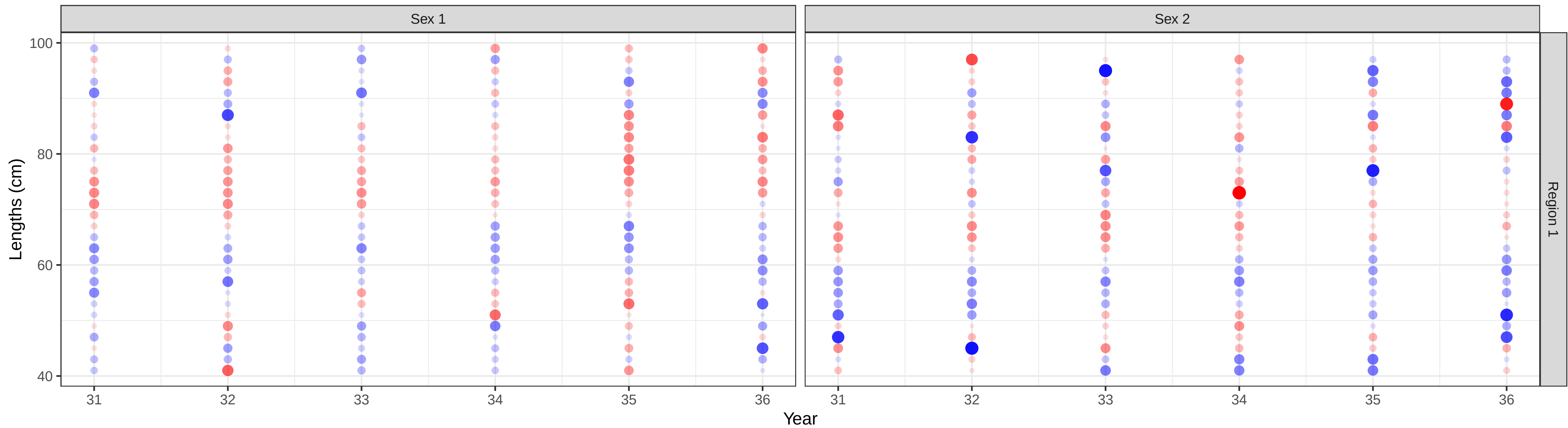
Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0 ● 2.5



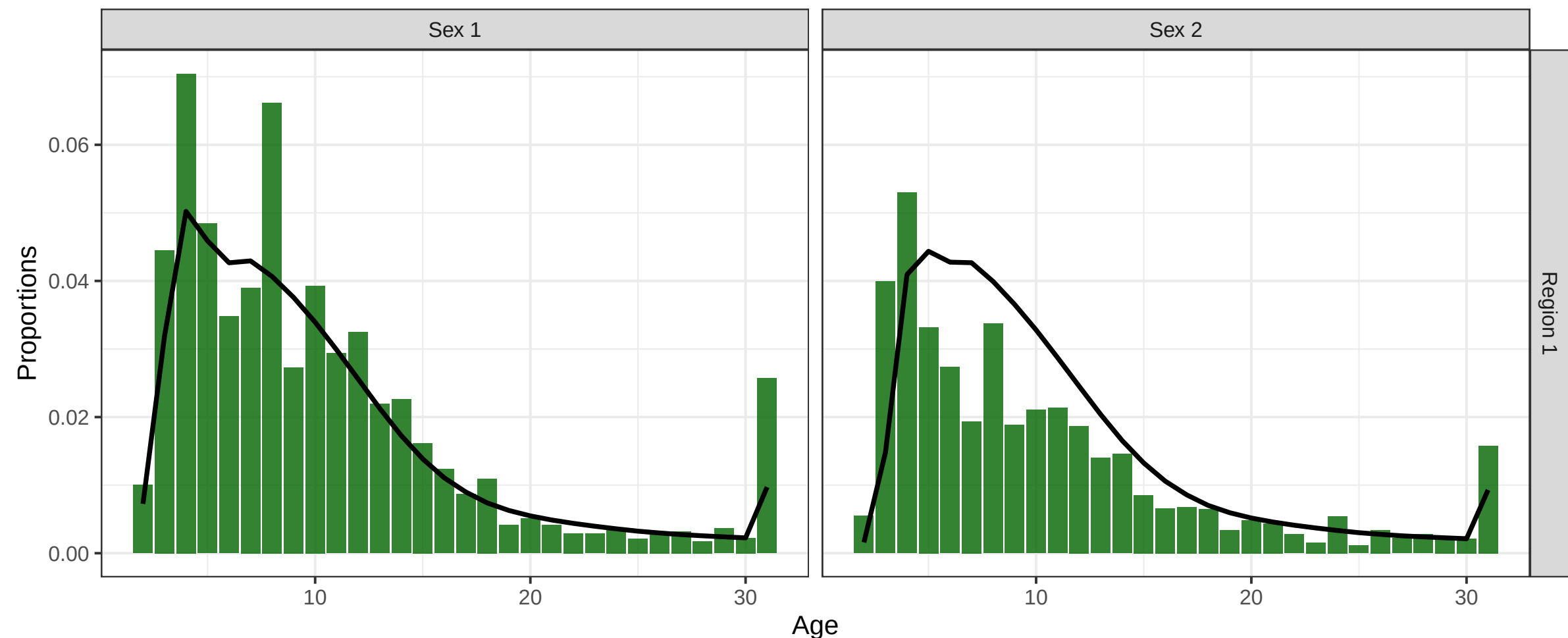
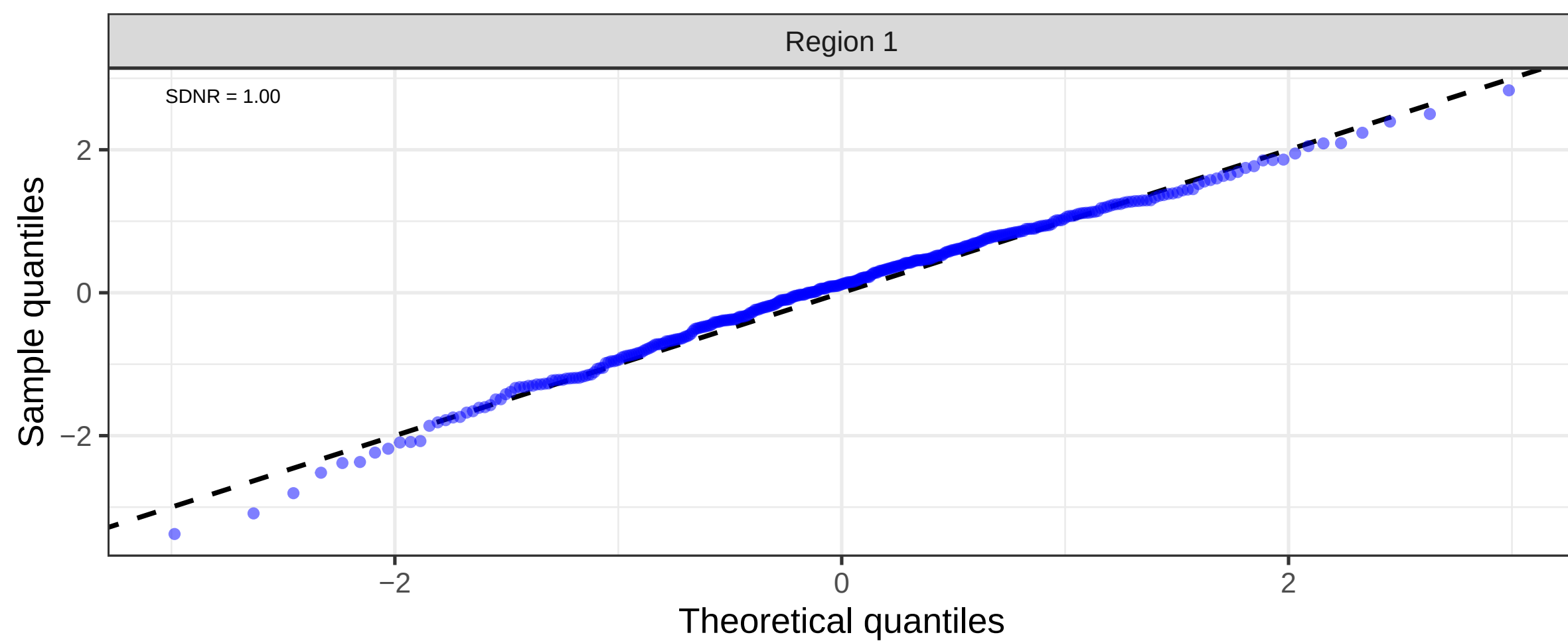
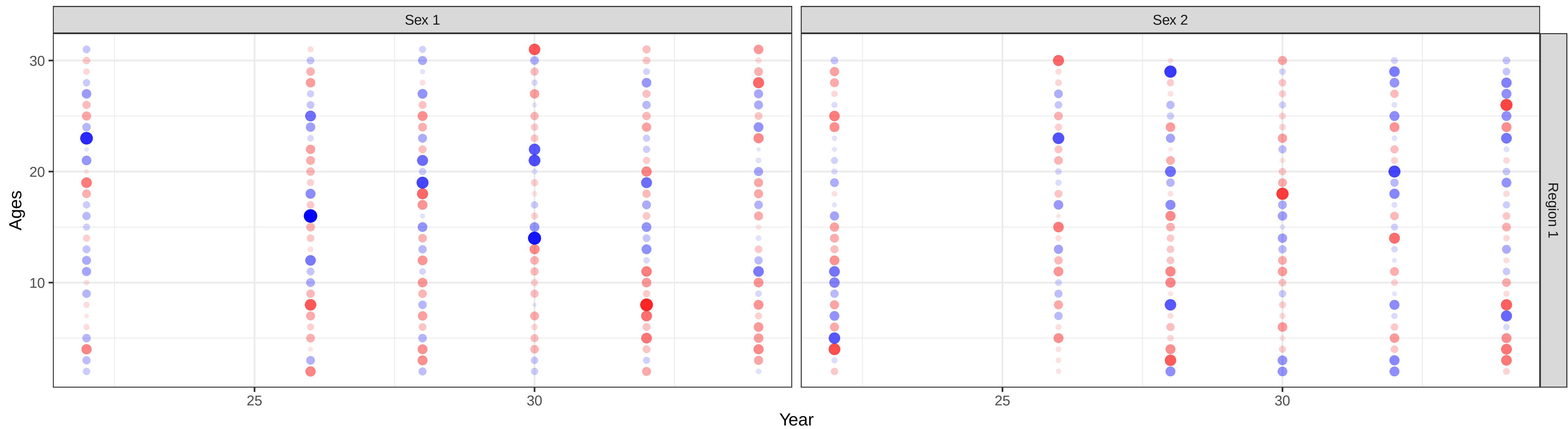


abs(Resid) 1 2 3 Sign ● Neg ● Pos

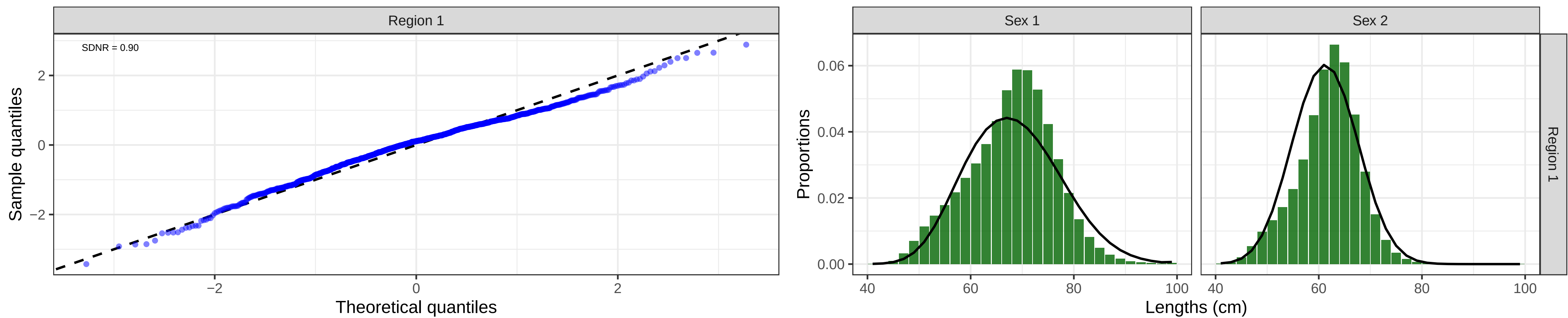
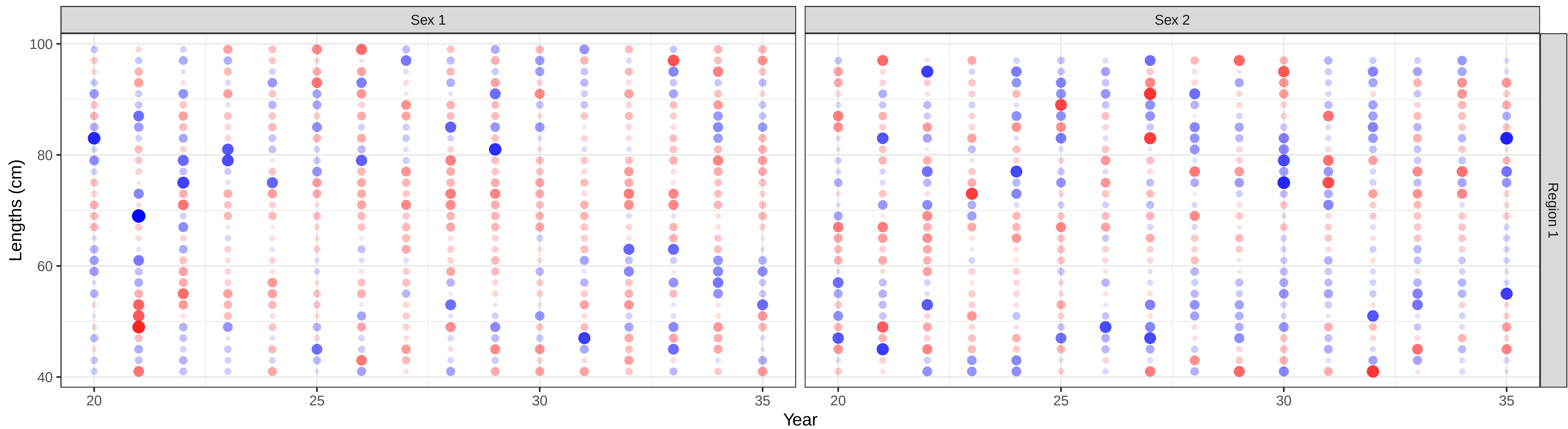




abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos

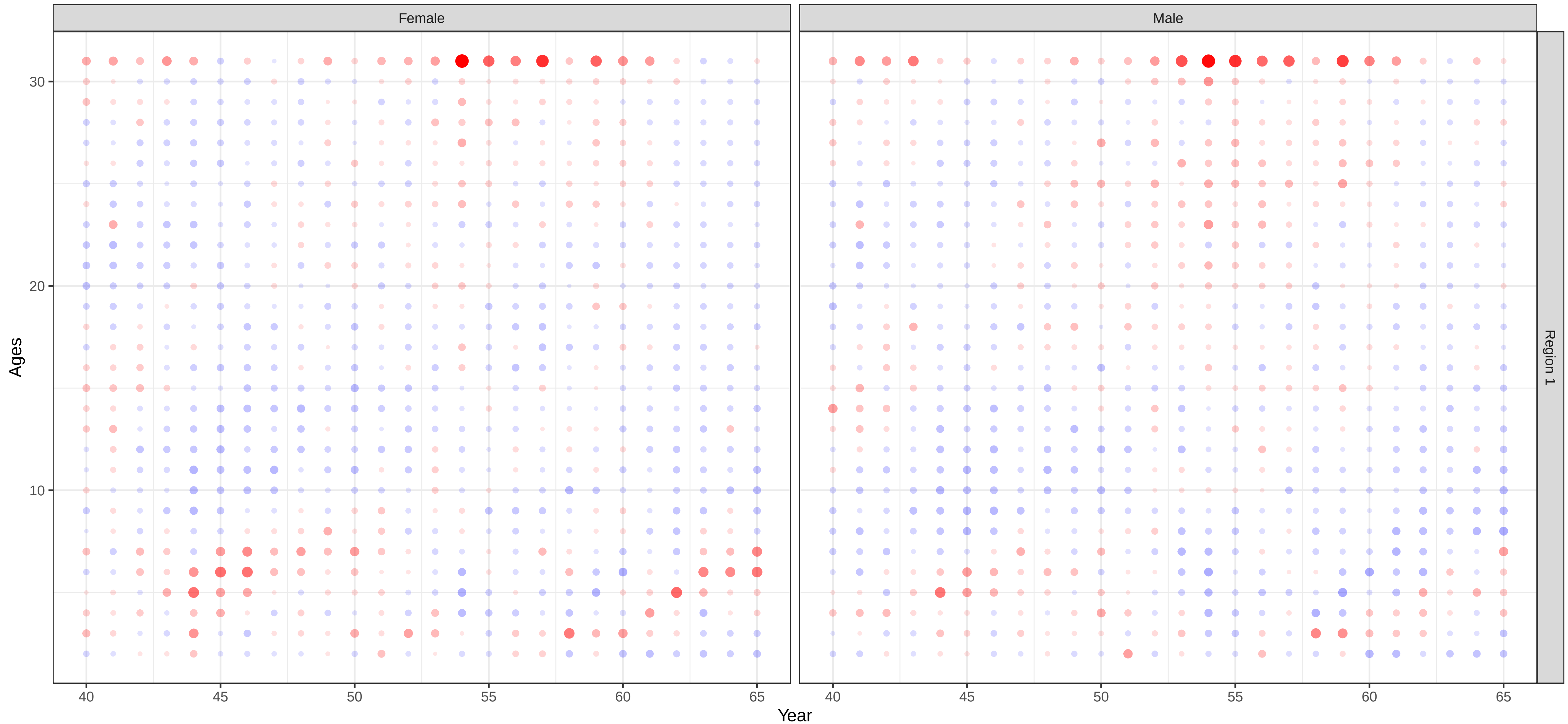


abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos



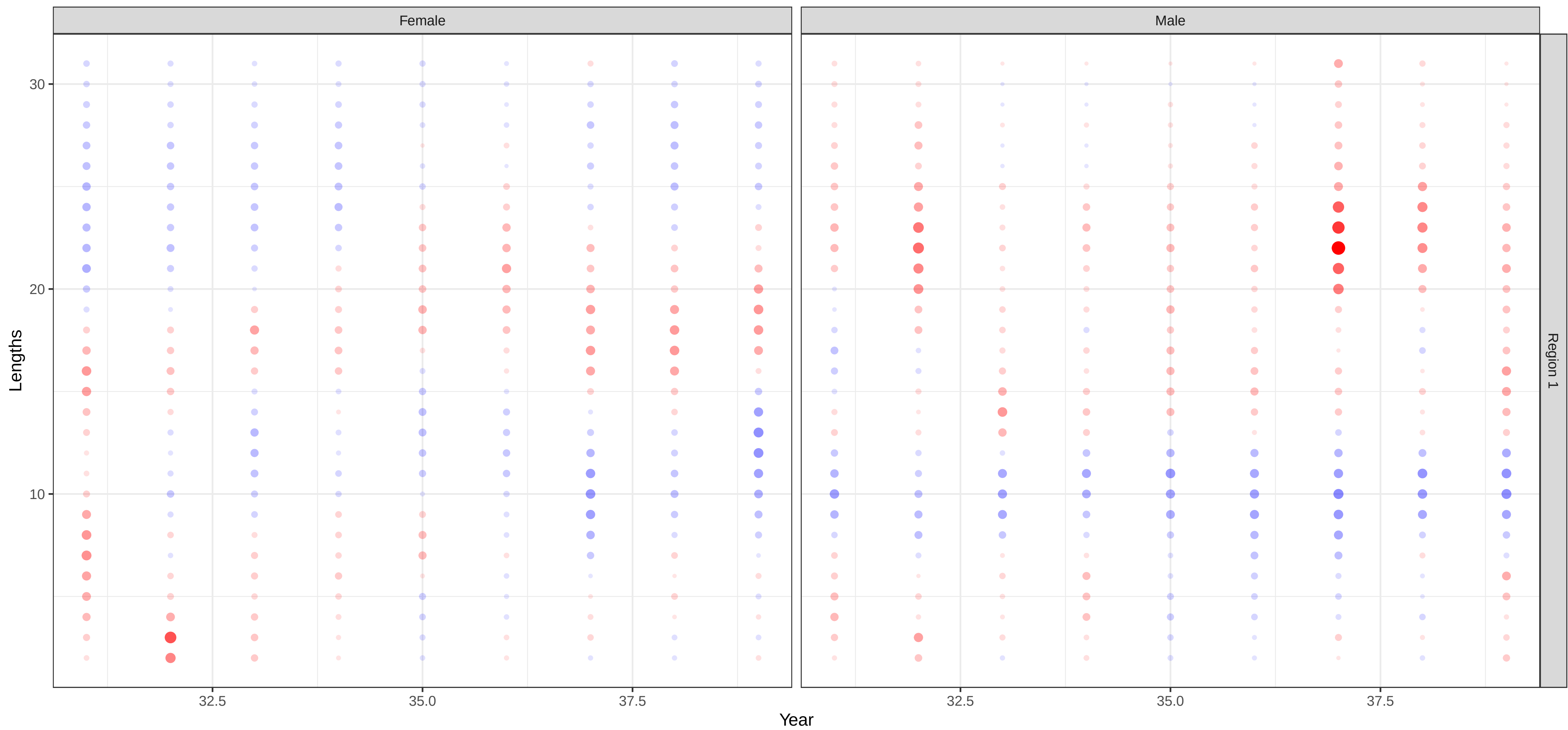
Fixed Gear Fishery Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0



Fixed Gear Fishery Length Compositions

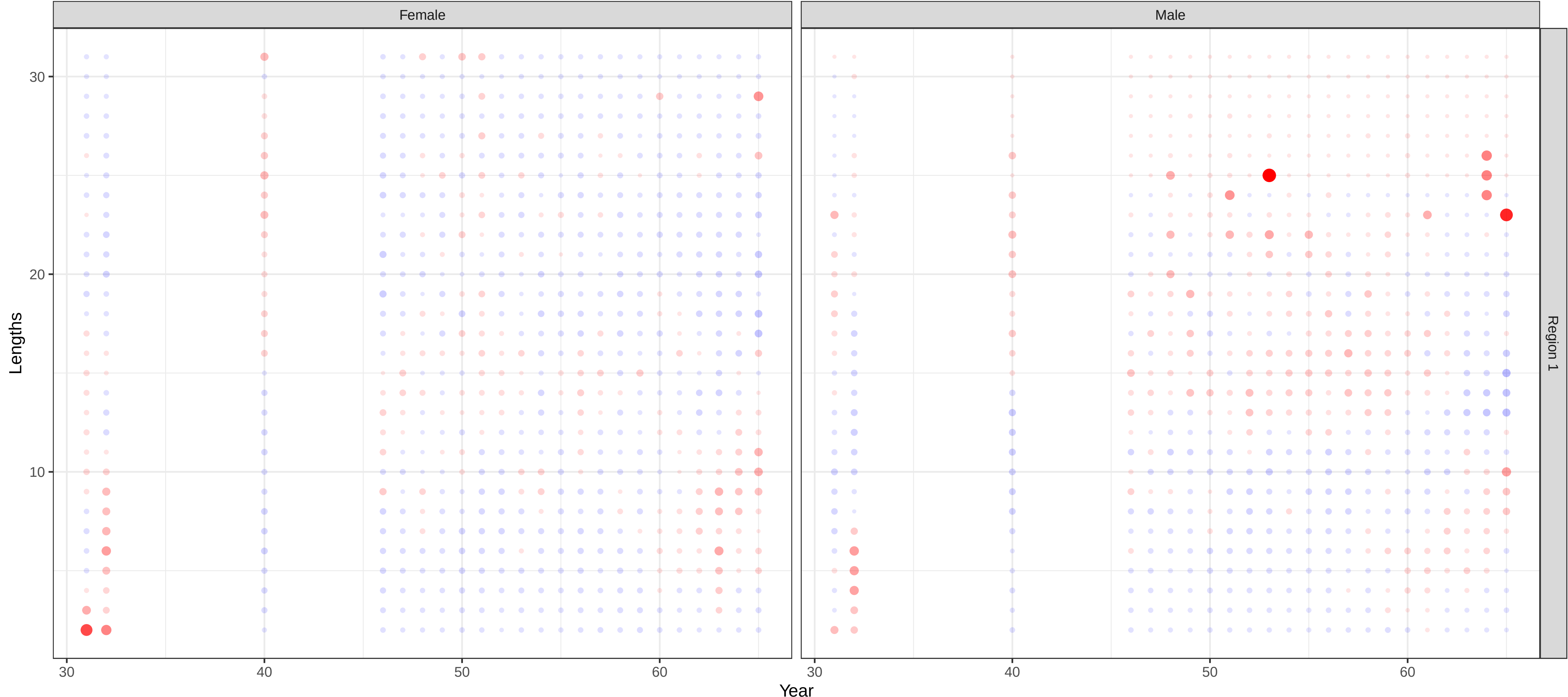
Sign ● Neg ● Pos abs(Resid) ● 0.0 ● 0.5 ● 1.0 ● 1.5 ● 2.0



Trawl Fishery Length Compositions

Sign ● Neg ● Pos

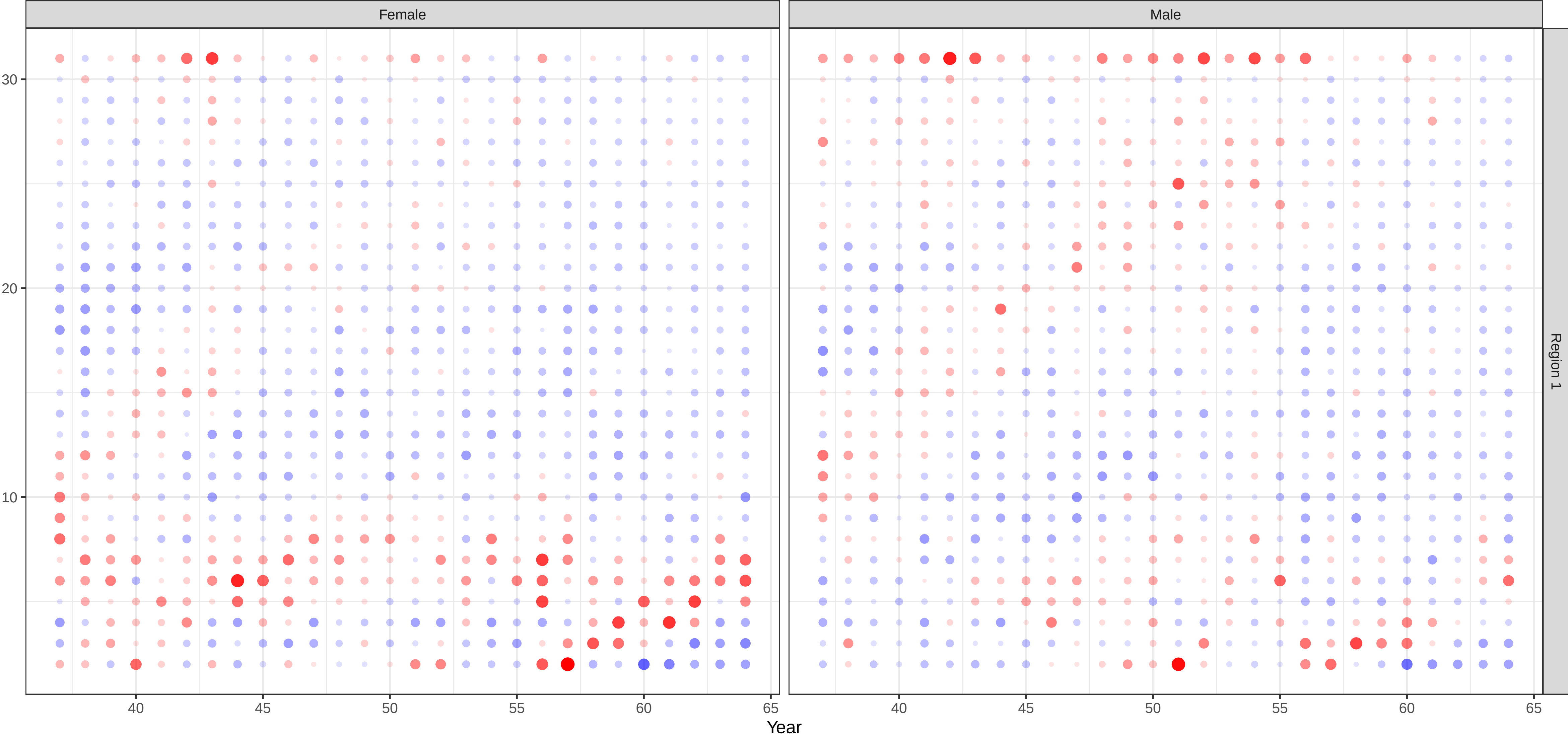
abs(Resid) ● 0 ● 2 ● 4
 ● 1 ● 3 ● 5



LLS Age Compositions

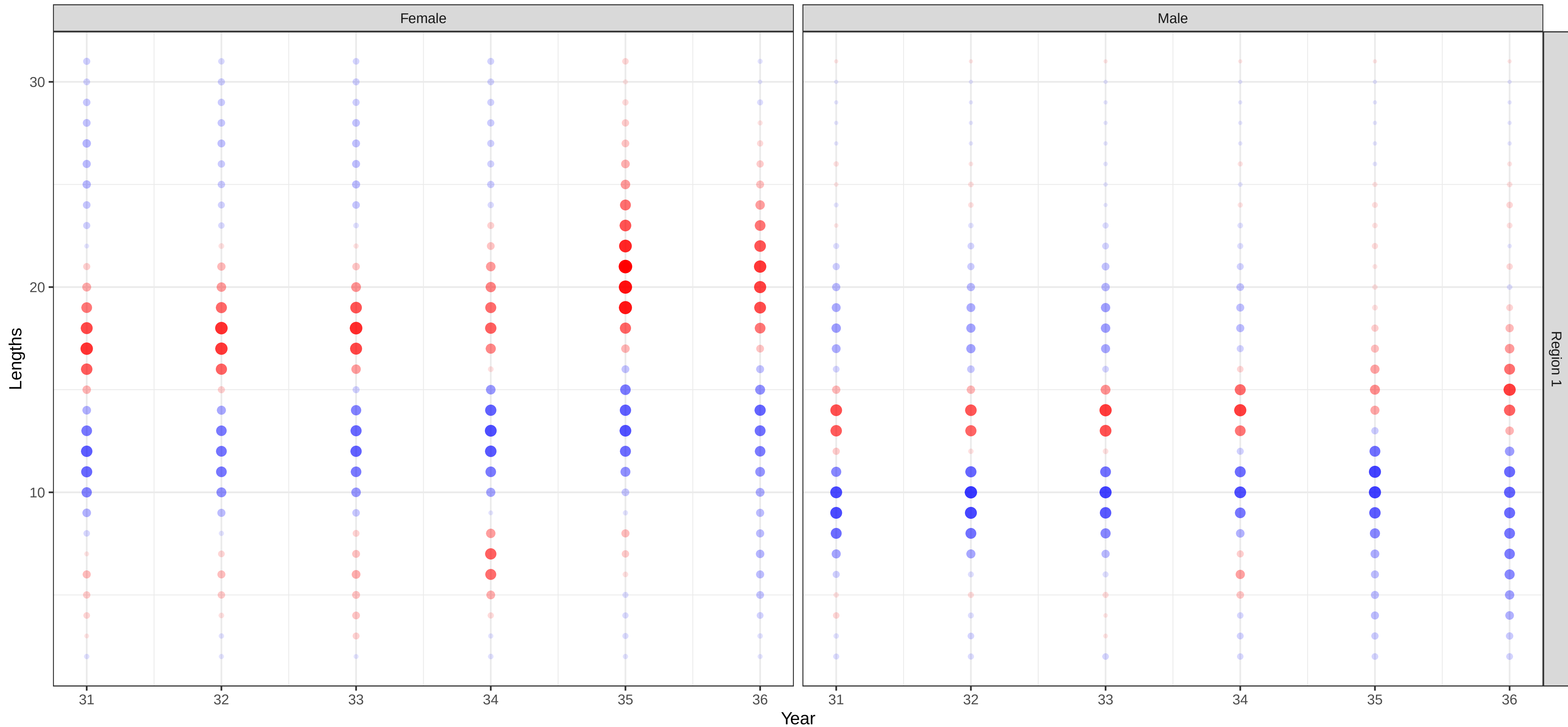
Sign ● Neg ● Pos

abs(Resid) ● 1 ● 2



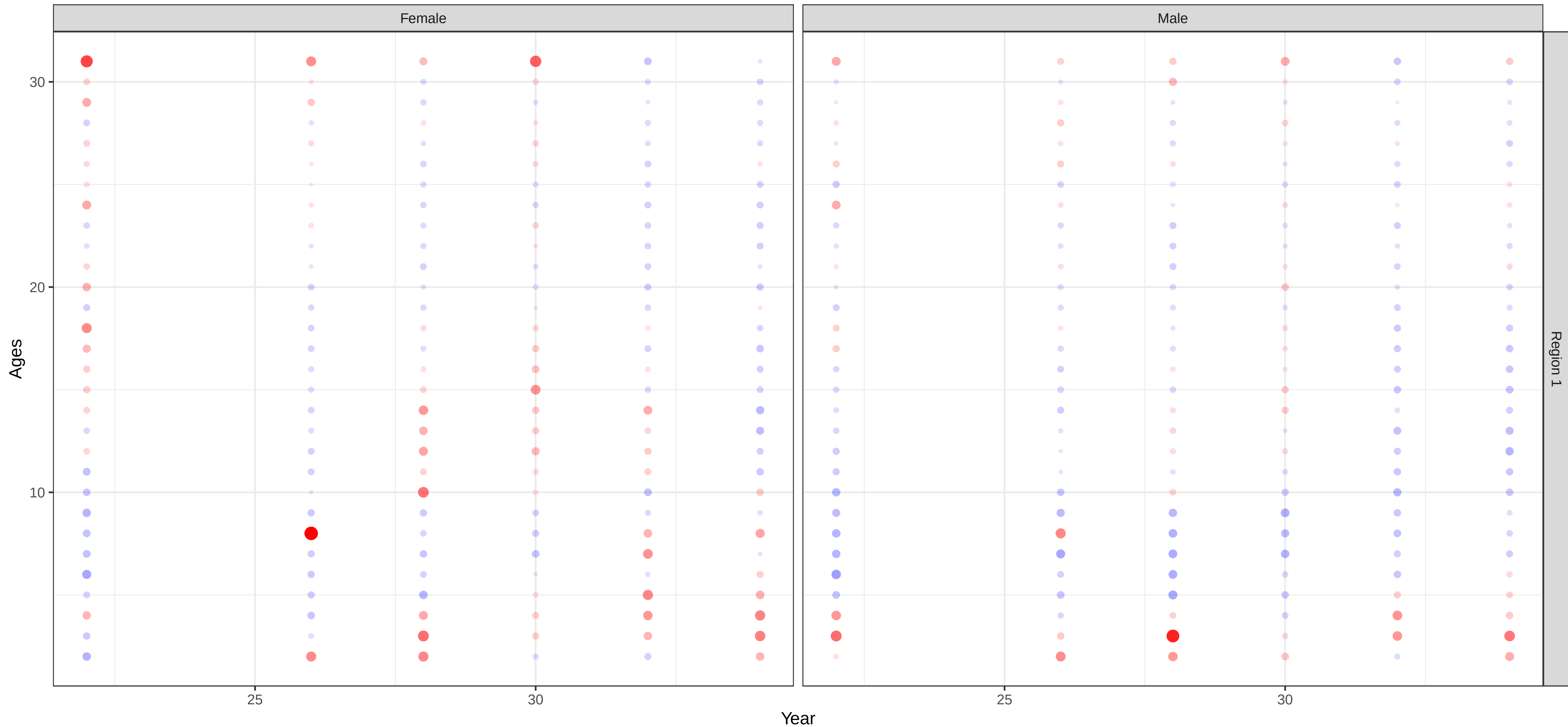
LLS Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.0 ● 0.5 ● 1.0



JPN LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0



JPN LLS Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.0 ● 0.5 ● 1.0 ● 1.5 ● 2.0

